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Marketing

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# Median-of-Means Two-Stage Least Squares Regression

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The views stated in this thesis are those of the author and not necessarily those of the supervisor, second assessor, Erasmus School of Economics or Erasmus University Rotterdam.

## Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method. All theoretical results in this thesis are formalised and machine-checked in Lean 4 using Mathlib; the proofs depend only on the three standard axioms of classical mathematics.

## 1 Introduction

The just-identified instrumental variable (IV) estimator is a ratio of two sample means. This makes it fragile in two distinct ways: the numerator is a sample mean of a possibly heavy-tailed product, and the denominator can be close to zero. The effects of large outliers on IV estimation was studied by Young (2022), where 1309 IV regressions in 30 papers were examined and it was shown that high leverage and non-iid errors are pervasive. This distorts IV tests and rarely rejects the OLS estimate. Additionally, F-statistics based on pre-tests become largely uninformative about size and bias.

In the presence of heavy tails, the sample mean is no longer an efficient measure of central tendency. As such, the quality of standard regression estimation techniques, such as OLS, can degrade significantly when this is the case. Under only a finite-variance assumption, the empirical mean’s deviation from the true mean scales as  $|\hat{\mu} - \mu| \lesssim \frac{1}{\sqrt{n\delta}}$  with confidence  $1 - \delta$ , whereas sub-Gaussian estimators achieve the exponentially better rate  $|\hat{\mu} - \mu| \lesssim \sqrt{\ln(1/\delta)/n}$ .

discuss about estimators that solve this and cite Devroye, Lerasle, Lugosi and Oliveira (2016)

A technique that has gained traction in the machine learning literature is the Median-of-Means (MoM) estimator (Lecué & Lerasle, 2020), which attains sub-Gaussian deviation bounds under nothing more than a finite variance. The foundational studies of this method were given by Nemirovskij and Yudin (1983), Jerrum, Valiant and Vazirani (1986), and Alon, Matias and Szegedy (1996). A comprehensive survey and analysis of the MoM estimator was given by Lugosi and Mendelson (2019)

MoM has been carried into OLS estimation, but not into instrumental variable estimation; MoM is defined for a mean, whereas the IV estimand is a ratio of means. Since there is no canonical way to take a median of a ratio, novel methods have to be constructed that, not only estimate it, but also give the most preferable theoretical guarantees, and empirical results.

We propose and analyse two MoM-based IV estimators. The Ratio-of-Medians (RoM) estimator takes the coordinate-wise median of the block cross-moments and then divides; the Median-of-Ratios (MoR) estimator divides within each block and then takes the median of the block-level IV estimates. Both of these estimators are Sub-Gaussian, but the RoM consistently shows better empirical results throughout the simulations.

The robustness is not free. In standard mean estimation MoM costs constants, but in ratio estimation it costs a per-block sample size floor. That is, in order to guarantee a Sub-Gaussian estimator with a certain probability, a minimum per-block sample size is needed. MoM-type IV estimators pay for tail robustness by spending instrument strength.

In order to be able to conduct inference under the MoM framework, we construct a median-of-means Anderson-Rubin (MoM-AR) test whose size is bounded by  $\delta$  in finite samples under the

same assumptions, and show that the confidence set obtained by inverting it can be computed exactly in closed form from at most  $2k$  candidate endpoints. This test depends on an unknown scale parameter  $\sigma$  that has to be estimated. We therefore construct a self-normalised Anderson-Rubin test which cancels the unknown  $\sigma$  instead of estimating it or treating it as known.

discuss size and power costs of tests

One or two sentences on the simulation evidence: robustness is nearly free under Gaussian errors, clearly beneficial at  $t(2.1)$ , and the  $\delta$ -sweep visibly reproduces the theoretical shapes.

Then the two applications, and make the pairing the point: Angrist and Krueger (1991) is the weak-instrument case and Angrist and Evans (1998) the strong one, so together they show the strength condition biting and not biting. Own the partition-sensitivity caveat here (Table 3) rather than letting a reader find it.

Finally, All theoretical results are formalised and machine-checked in Lean 4 using the Mathlib library, so that the measure-theoretic and probabilistic foundations are inherited rather than assumed.

The remainder of the paper is organised as follows. Section 3

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; Sections 4 and 5 introduce the RoM and MoR estimators; Section 6 develops the inference procedures; Section 7.3 reports the simulation evidence; and Section 9.1

fix – point at the case studies section, which currently has no label

## 2 Preliminaries

### 2.1 Setup and Assumptions

Let  $(Y_i, X_i, Z_i)_{i=1}^n$  be an i.i.d. random sample from a joint distribution on  $\mathbb{R}^3$  satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \quad (1)$$

where  $Y_i$  is the variable of interest,  $X_i$  is an endogenous regressor, and  $Z_i$  is the instrumental variable (IV). Additionally,  $\beta \in \mathbb{R}$  is the structural parameter of interest and  $\varepsilon_i$  is an unobserved error term.

For any random quantity  $f$ , we write  $\mu_f := \mathbb{E}[f]$  and  $\sigma_f^2 := \text{Var}(f)$  for its mean and variance. All population quantities below are instances of this scheme; for example,  $\mu_X = \mathbb{E}[X_i]$ ,  $\mu_{ZX} = \mathbb{E}[Z_i X_i]$ , and  $\sigma_{ZX}^2 = \text{Var}(Z_i X_i)$ .

We impose the following conditions.

(A1) **Exogeneity.**  $\mu_{Z\varepsilon} = 0$ .

(A2) **Relevance.**  $\mu_{ZX} \neq 0$ .

(A3) **Finite second moments.**  $\sigma_{ZX}^2, \sigma_{ZY}^2, \sigma_{Z\varepsilon}^2 < \infty$ .

By (A1) and (A2), the structural parameter is identified as  $\beta = \mu_{ZY}/\mu_{ZX}$ .

MoM  
can also  
be in  
sub-  
script

**Remark 2.1** (Moment conditions). *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products  $Z_i Y_i$ ,  $Z_i X_i$ , and  $Z_i \varepsilon_i$ . Thus, no assumptions are imposed on the marginal distributions of  $Y_i$ ,  $X_i$ ,  $Z_i$ , or  $\varepsilon_i$ .*

**Remark 2.2** (Relationship between variances). *The three variances in (A3) are related. Since  $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$ , Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}. \quad (2)$$

*To see this, set  $U = Z_i Y_i$  and  $V = \beta Z_i X_i$ , so that  $U - V = Z_i \varepsilon_i$ . By Minkowski's inequality for  $p = 2$ ,*

$$\sqrt{\text{Var}(U - V)} \leq \sqrt{\text{Var}(U)} + \sqrt{\text{Var}(V)},$$

*which gives  $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}$ .*

## 2.2 Sub-Gaussian Estimators

**Definition 2.3** (Sub-Gaussian Estimators). *Let  $(X_i)_{i=1}^n$  be a random sample with mean  $\mu_X$  and variance  $\sigma_X^2$ , and let  $\hat{\mu}_X$  denote an estimator for  $\mu_X$ . For some constant  $L > 0$ ,  $\hat{\mu}_X$  is said to be a sub-Gaussian estimator for  $\mu_X$  for a given confidence level  $\delta$  if*

$$\mathbb{P} \left[ |\hat{\mu}_X - \mu_X| > L \sigma_X \sqrt{\frac{\ln(1/\delta)}{n}} \right] \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (3)$$

## 2.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye et al. (2016) and Lugosi and Mendelson (2019).

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### Algorithm 1: Scalar Median-of-Means Estimator

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**Input:** Sample  $(X_i)_{i=1}^n$ , number of blocks  $k$

**Output:** Estimate  $\hat{\mu}_{\text{MoM}}$

Partition  $(X_i)_{i=1}^n$  into  $k$  blocks  $B_1, \dots, B_k$  of size  $m = \lfloor n/k \rfloor$ ;

**for**  $j = 1, \dots, k$  **do**

    Compute the block mean  $\hat{\mu}_X^{(j)} = \frac{1}{m} \sum_{i \in B_j} X_i$ ;

**return**  $\hat{\mu}_{\text{MoM}} = \text{med}(\hat{\mu}_X^{(1)}, \dots, \hat{\mu}_X^{(k)})$ ;

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**Theorem 2.4** (MoM concentration). *Let  $(X_i)_{i=1}^n$  be i.i.d. with mean  $\mu_X$  and variance  $\sigma_X^2 < \infty$ . Let  $k$  be a positive integer,  $m = \lfloor n/k \rfloor$ , and  $\hat{\mu}_{\text{MoM}}$  the MoM estimator of  $\mu_X$ . Then*

$$\mathbb{P} \left[ |\hat{\mu}_{\text{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}} \right] \leq e^{-k/8}. \quad (4)$$

*Setting  $k = \lceil 8 \ln(1/\delta) \rceil$  for  $\delta \in (0, 1)$ , this becomes*

$$\mathbb{P} \left[ |\hat{\mu}_{\text{MoM}} - \mu_X| > \sigma_X \sqrt{\frac{32 \ln(1/\delta)}{n}} \right] \leq \delta, \quad (5)$$

*which is sub-Gaussian in the sense of definition 2.3 with  $L = \sqrt{32}$ .*

Think  
about  $\mu$   
notation

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 $n = km$

*Proof.* The proof proceeds in three steps.

**Step 1 (Chebyshev on each block).** Each block mean satisfies  $\mathbb{E}[\bar{X}^{(j)}] = \mu_X$  and  $\text{Var}(\bar{X}^{(j)}) = \sigma_X^2/m$ . By Chebyshev's inequality with threshold  $2\sigma_X/\sqrt{m}$ ,

$$\mathbb{P}\left[\left|\bar{X}^{(j)} - \mu_X\right| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \frac{\sigma_X^2/m}{4\sigma_X^2/m} = \frac{1}{4}.$$

**Step 2 (Counting argument for the median).** Define  $\zeta^{(j)} = \mathbf{1}\{|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}\}$ . By Step 1,  $\mathbb{E}[\zeta^{(j)}] \leq 1/4$ . Since the blocks are disjoint subsets of an i.i.d. sample, the  $(\zeta^{(j)})_{j=1}^k$  are independent.

If  $|\hat{\mu}_{\text{MoM}} - \mu_X| > 2\sigma_X/\sqrt{m}$ , the median exceeds  $\mu_X + 2\sigma_X/\sqrt{m}$  or falls below  $\mu_X - 2\sigma_X/\sqrt{m}$ . In either case, at least  $k/2$  of the block means satisfy  $|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}$ , since the median of  $k$  numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P}\left[|\hat{\mu}_{\text{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta^{(j)} \geq \frac{k}{2}\right].$$

**Step 3 (Hoeffding's inequality).** The sum  $\sum_{j=1}^k \zeta^{(j)}$  consists of  $k$  independent random variables with  $\zeta^{(j)} \in [0, 1]$  and  $\mathbb{E}[\sum_j \zeta^{(j)}] \leq k/4$ . For  $\sum_j \zeta^{(j)}$  to reach  $k/2$ , it must exceed its mean by at least  $k/2 - k/4 = k/4$ . By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P}\left[\sum_{j=1}^k \zeta^{(j)} - \mathbb{E}\left[\sum_j \zeta^{(j)}\right] \geq \frac{k}{4}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting  $k = \lceil 8 \ln(1/\delta) \rceil$ .  $\square$

### 3 The Standard Instrumental Variable Estimator

**Definition 3.1** (Standard IV estimator). *Consider the structural equation (1). The standard IV estimator  $\hat{\beta}_{\text{IV}}$  is defined by*

$$\hat{\beta}_{\text{IV}} = \frac{\hat{\mu}_{ZY}}{\hat{\mu}_{ZX}},$$

**Lemma 3.2** (Error decomposition). *Under (1) and (A1),*

$$\hat{\beta}_{\text{IV}} - \beta = \frac{\hat{\mu}_{Z\varepsilon}}{\hat{\mu}_{ZX}}, \tag{6}$$

where  $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$ .

*Proof.* Multiply both sides of  $Y_i = \beta X_i + \varepsilon_i$  by  $Z_i$  and average:

$$\hat{\mu}_{ZY} = \frac{1}{n} \sum_{i=1}^n Z_i(\beta X_i + \varepsilon_i) = \beta \hat{\mu}_{ZX} + \hat{\mu}_{Z\varepsilon}.$$

Rearranging and dividing by  $\hat{\mu}_{ZX}$  gives  $\hat{\beta}_{\text{IV}} = \beta + \hat{\mu}_{Z\varepsilon}/\hat{\mu}_{ZX}$ , which is (6). By (A1),  $\mathbb{E}[Z_i \varepsilon_i] = 0$ , so  $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$  by linearity of expectation.  $\square$

**Theorem 3.3** (Standard IV concentration). *Assume (A1)–(A3). For any  $\delta \in (0, 1)$ , if*

$$n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad (7)$$

then

$$\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > \frac{2\sqrt{2}\sigma_{Z\epsilon}}{|\mu_{ZX}|\sqrt{\delta n}}\right] \leq \delta. \quad (8)$$

*Proof.* By Lemma 3.2, the estimation error equals the ratio  $\hat{\mu}_{Z\epsilon}/\hat{\mu}_{ZX}$ . To bound this ratio we need two things simultaneously: the denominator must be bounded away from zero, and the numerator must be small. We control each requirement with a separate probability event and then combine them via the union bound.

**Step 1 (Two-event decomposition).** For a denominator threshold  $D \in (0, |\mu_{ZX}|)$  and an error threshold  $t > 0$ , both to be chosen below, define

$$A(D) = \{|\hat{\mu}_{ZX}| \geq D\}, \quad B(D, t) = \{|\hat{\mu}_{Z\epsilon}| \leq tD\}.$$

On  $A(D) \cap B(D, t)$ , the denominator satisfies  $|\hat{\mu}_{ZX}| \geq D$  by definition of  $A(D)$ , so

$$\left|\hat{\beta}_{IV} - \beta\right| = \frac{|\hat{\mu}_{Z\epsilon}|}{|\hat{\mu}_{ZX}|} \leq \frac{tD}{D} = t.$$

Hence  $\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A(D)^c] + \mathbb{P}[B(D, t)^c]$  by the union bound. For the remainder of this proof we set  $D = |\mu_{ZX}|/2$ , the symmetric choice that yields the cleanest constants.

**Step 2 (Chebyshev bounds).** We bound each failure probability in turn.

The failure event  $A^c = \{|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2\}$  implies a large deviation of  $\hat{\mu}_{ZX}$  from its mean. Specifically, if  $|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2$  then, by the reverse triangle inequality,

$$|\hat{\mu}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - |\hat{\mu}_{ZX}| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence  $A^c \subseteq \{|\hat{\mu}_{ZX} - \mu_{ZX}| > |\mu_{ZX}|/2\}$ . Since  $\text{Var}(\hat{\mu}_{ZX}) = \sigma_{ZX}^2/n$ , Chebyshev's inequality applied with threshold  $|\mu_{ZX}|/2$  gives

$$\mathbb{P}[A^c] \leq \frac{\sigma_{ZX}^2/n}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2}.$$

The failure event is  $B^c = \{|\hat{\mu}_{Z\epsilon}| > t|\mu_{ZX}|/2\}$ . Since  $\mathbb{E}[\hat{\mu}_{Z\epsilon}] = 0$  and  $\text{Var}(\hat{\mu}_{Z\epsilon}) = \sigma_{Z\epsilon}^2/n$ , Chebyshev's inequality gives

$$\mathbb{P}[B^c] \leq \frac{\sigma_{Z\epsilon}^2/n}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\epsilon}^2}{nt^2\mu_{ZX}^2}.$$

**Step 3 (Allocate the error budget and solve for  $t$ ).** We require the total failure probability  $\mathbb{P}[A^c] + \mathbb{P}[B^c]$  to be at most  $\delta$ , and we allocate half the budget to each event.

Budget for  $A^c$ . Setting  $\mathbb{P}[A^c] \leq \delta/2$  requires

$$\frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2} \leq \frac{\delta}{2} \iff n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2},$$

which is exactly condition (7).

Setting  $\mathbb{P}[B^c] \leq \delta/2$  and solving for  $t$ :

$$\frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2} = \frac{\delta}{2} \implies t = \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}.$$

By the union bound,

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta. \quad \square$$

**Remark 3.4** (Polynomial dependence on  $\delta$ ). *The bound (8) contains the factor  $1/\sqrt{\delta}$ , which is polynomial in  $1/\delta$ . Devroye et al. (2016) show that this is unavoidable: for any estimator formed from the empirical mean of i.i.d. observations with only finite variance, the  $1/\sqrt{\delta}$  rate cannot be improved without additional distributional assumptions. The MoM-based estimators in Sections 4–5 circumvent this by operating on block means.*

**Remark 3.5** (The symmetric threshold is not optimal). *The choice  $D = |\mu_{ZX}|/2$  was made for simplicity. It splits the distance between 0 and  $|\mu_{ZX}|$  evenly, but this is not the split that minimises the error bound for given  $n$  and  $\delta$ . Section 3.1 shows that treating  $D$  as a free parameter and optimising it reduces the instrument strength constant from 8 to 1 and improves the error constant by a factor of  $2\sqrt{2}$ .*

### 3.1 Optimised Threshold

The proof of Theorem 3.3 introduced a general threshold  $D$  but then fixed it symmetrically. Here we keep  $D$  free and choose it to minimise the resulting error bound, at the cost of a more involved Step 3.

**Theorem 3.6** (Standard IV concentration, optimised threshold). *Assume (A1)–(A3). Define*

$$\rho_{\text{IV}} = \frac{\sigma_{ZX}^2}{\delta n \mu_{ZX}^2}.$$

*For any  $\delta \in (0, 1)$ , if  $\rho_{\text{IV}} < 1$  (i.e.,  $n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$ ), then the choice  $D^* = (1 - \rho_{\text{IV}}^{1/3})|\mu_{ZX}|$  is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}(1 - \rho_{\text{IV}}^{1/3})^{3/2}}\right] \leq \delta. \quad (9)$$

*Proof.* Steps 1 and 2 are identical to those of Theorem 3.3 but with  $D$  left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}| - D)^2}, \quad \mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Requiring their sum to be at most  $\delta$  gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2 D^2} \leq \delta. \quad (10)$$

**Step 3 (Optimise  $D$ , then solve for  $t$ ).** Write  $D = \eta |\mu_{ZX}|$  with  $\eta \in (0, 1)$ . Dividing (10) through by  $\delta$  and substituting  $\rho_{IV} = \sigma_{ZX}^2 / (\delta n \mu_{ZX}^2)$ :

$$\frac{\rho_{IV}}{(1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2 \eta^2 t^2} \leq 1.$$

Solving for  $t^2$ :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \quad f(\eta) = \eta^2 \left( 1 - \frac{\rho_{IV}}{(1 - \eta)^2} \right).$$

Minimising  $t$  is equivalent to maximising  $f$  over  $\eta \in (0, 1)$ . Differentiating and setting  $f'(\eta) = 0$ :

$$f'(\eta) = 2\eta \left( 1 - \frac{\rho_{IV}}{(1 - \eta)^2} \right) - \frac{2\rho_{IV}\eta^2}{(1 - \eta)^3} = 0.$$

Dividing through by  $2\eta > 0$ :

$$1 - \frac{\rho_{IV}}{(1 - \eta)^2} - \frac{\rho_{IV}\eta}{(1 - \eta)^3} = 0 \iff (1 - \eta)^3 - \rho_{IV}(1 - \eta) - \rho_{IV}\eta = 0 \iff (1 - \eta)^3 = \rho_{IV}.$$

Hence  $\eta^* = 1 - \rho_{IV}^{1/3}$ , which lies in  $(0, 1)$  if and only if  $\rho_{IV} < 1$ , the stated instrument strength condition. Substituting  $(1 - \eta^*)^2 = \rho_{IV}^{2/3}$ :

$$f(\eta^*) = (1 - \rho_{IV}^{1/3})^2 \left( 1 - \frac{\rho_{IV}}{\rho_{IV}^{2/3}} \right) = (1 - \rho_{IV}^{1/3})^2 (1 - \rho_{IV}^{1/3}) = (1 - \rho_{IV}^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n} (1 - \rho_{IV}^{1/3})^{3/2}},$$

and the union bound gives  $\mathbb{P} \left[ \left| \hat{\beta}_{IV} - \beta \right| > t^* \right] \leq \delta$ . □

**Remark 3.7** (Comparison with the symmetric bound). As  $\rho_{IV} \rightarrow 0$  (strong instruments or large  $n$ ), the correction factor  $(1 - \rho_{IV}^{1/3})^{-3/2} \rightarrow 1$  and (9) approaches  $\sigma_{Z\varepsilon} / (|\mu_{ZX}| \sqrt{\delta n})$ , a factor of  $2\sqrt{2} \approx 2.83$  tighter than (8). As  $\rho_{IV} \rightarrow 1$  (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 3.3:  $\rho_{IV} < 1$  versus  $\rho_{IV} < 1/8$ .

## 3.2 Cantelli Improvement

The Chebyshev bound applied to  $A(D)^c$  in the proofs above is two-sided. However,  $A(D)^c = \{|\hat{\mu}_{ZX}| < D\}$  is a one-sided failure event: only a *downward* deviation of  $\hat{\mu}_{ZX}$  from  $\mu_{ZX}$  causes

the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

**Lemma 3.8** (Cantelli's inequality). *Let  $W$  be a random variable with  $\mathbb{E}[W] = \mu$  and  $\text{Var}(W) = \sigma^2 < \infty$ . For any  $\tau > 0$ ,*

$$\mathbb{P}[W - \mu \leq -\tau] \leq \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

*This is strictly smaller than the Chebyshev bound  $\sigma^2/\tau^2$  for all  $\tau > 0$ .*

**Theorem 3.9** (Standard IV concentration, Cantelli improvement). *Assume (A1)–(A3) and  $\mu_{ZX} > 0$ . Write  $\gamma_{\text{IV}} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$  and define*

$$h(\eta) = \eta^2 \frac{\delta(\gamma_{\text{IV}} + (1 - \eta)^2) - \gamma_{\text{IV}}}{\gamma_{\text{IV}} + (1 - \eta)^2}, \quad \eta \in (0, 1). \quad (11)$$

*If*

$$n > \frac{(1 - \delta)\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad \text{equivalently,} \quad \gamma_{\text{IV}} < \frac{\delta}{1 - \delta}, \quad (12)$$

*then  $h(\eta) > 0$  on a non-empty interval and the estimator satisfies*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{nh(\eta_{\text{IV}}^*)}}\right] \leq \delta, \quad (13)$$

*where  $\eta_{\text{IV}}^*$  is the maximiser of  $h$  over its feasible domain.*

*Proof.* We use the same two-event framework with  $D = \eta\mu_{ZX}$  kept general.

**Step 1 (Two-event decomposition).** As in Theorem 3.3, define  $A(D) = \{\hat{\mu}_{ZX} \geq D\}$  and  $B(D, t) = \{|\hat{\mu}_{Z\varepsilon}| \leq tD\}$ . On  $A(D) \cap B(D, t)$ ,  $|\hat{\beta}_{\text{IV}} - \beta| \leq t$ .

**Step 2 (Cantelli on  $A^c$ , Chebyshev on  $B^c$ ).**

Since  $\mu_{ZX} > 0$  and  $D < \mu_{ZX}$ , the failure event satisfies

$$A(D)^c = \{\hat{\mu}_{ZX} < D\} \subseteq \{\hat{\mu}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 3.8 applies with  $W = \hat{\mu}_{ZX}$ ,  $\sigma^2 = \sigma_{ZX}^2/n$ , and  $\tau = \mu_{ZX} - D$ :

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{\sigma_{ZX}^2/n + (\mu_{ZX} - D)^2}.$$

The event  $B(D, t)^c = \{|\hat{\mu}_{Z\varepsilon}| > tD\}$  is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

**Step 3 (Constraint and optimization).** Setting the total failure probability to at most  $\delta$  and substituting  $D = \eta\mu_{ZX}$ ,  $\gamma_{\text{IV}} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$ :

$$\frac{\gamma_{\text{IV}}}{\gamma_{\text{IV}} + (1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2\eta^2\mu_{ZX}^2} \leq \delta.$$

Isolating the second term and solving for  $t^2$ :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{n\mu_{ZX}^2 h(\eta)},$$

where  $h(\eta)$  is defined in (11). The function  $h(\eta)$  is positive only when  $\delta(\gamma_{IV} + (1 - \eta)^2) > \gamma_{IV}$ , i.e.,  $(1 - \eta)^2 > \gamma_{IV}(1 - \delta)/\delta$ . The feasible region is non-empty if and only if  $\gamma_{IV} < \delta/(1 - \delta)$ , which is condition (12). Minimizing  $t$  over the feasible region by maximizing  $h$  and applying the union bound gives (13).  $\square$

**Remark 3.10** (When Cantelli helps). *Compared with the optimised Chebyshev condition ( $n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$ ), the Cantelli condition (12) is weaker by a factor of  $1/(1 - \delta)$ . At  $\delta = 0.05$  this ratio is  $1/0.95 \approx 1.05$ , so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at  $\delta = 0.5$  the factor reaches 2. For the standard IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section 5.2), Cantelli extends the valid range of the instrument strength parameter from  $\gamma < 1/4$  to  $\gamma < 1/3$ , a qualitatively more substantial gain.*

**Remark 3.11** (No tractable closed form for  $\eta_{IV}^*$ ). *The first-order condition  $h'(\eta) = 0$  reduces to a quartic in  $\eta$  with no tractable closed-form solution. The optimised Chebyshev version (Theorem 3.6) yielded the clean form  $\eta^* = 1 - \rho_{IV}^{1/3}$  because the Chebyshev bound  $\sigma^2/\tau^2$  produces a simpler algebraic structure than the Cantelli bound  $\sigma^2/(\sigma^2 + \tau^2)$ . The bound (13) must therefore be evaluated numerically for specific parameter values.*

## 4 The Ratio-of-Medians (RoM) Estimator

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### Algorithm 2: Ratio-of-Medians Estimator

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**Input:** Sample  $(Y_i, X_i, Z_i)_{i=1}^n$ , number of blocks  $k$

**Output:** Estimate  $\hat{\beta}_{RM}$

Partition  $(Y_i, X_i, Z_i)_{i=1}^n$  into  $k$  blocks  $B_1, \dots, B_k$  of size  $m = \lfloor n/k \rfloor$ ;

**for**  $j = 1, \dots, k$  **do**

    Compute the block means  $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$  and  $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$ ;

Compute the coordinate-wise medians  $\tilde{\mu}_{ZY} = \text{med}(\hat{\mu}_{ZY}^{(1)}, \dots, \hat{\mu}_{ZY}^{(k)})$  and

$\tilde{\mu}_{ZX} = \text{med}(\hat{\mu}_{ZX}^{(1)}, \dots, \hat{\mu}_{ZX}^{(k)})$ ;

**return**  $\hat{\beta}_{RM} = \frac{\tilde{\mu}_{ZY}}{\tilde{\mu}_{ZX}}$ ;

---

geometric  
median

**Theorem 4.1** (RoM concentration). *Assume (A1)–(A3). Let  $\delta \in (0, 1)$ ,  $k = \lceil 8 \ln(2/\delta) \rceil$ , and  $m = \lfloor n/k \rfloor$ . If the instrument strength condition*

$$m > \frac{4\sigma_{ZX}^2}{\mu_{ZX}^2} \quad (14)$$

*holds, then*

$$\mathbb{P}\left[\left|\hat{\beta}_{RM} - \beta\right| > \frac{2(\sigma_{ZY} + |\beta| \sigma_{ZX})}{|\mu_{ZX}| \sqrt{m} - 2\sigma_{ZX}}\right] \leq \delta. \quad (15)$$

*Proof.* The proof applies the scalar MoM result (Theorem 2.4) to each coordinate separately and combines the results with a union bound.

**Step 1 (MoM on each coordinate).** By Theorem 2.4 applied to the sequences  $(Z_i Y_i)_{i=1}^n$  and  $(Z_i X_i)_{i=1}^n$ ,

$$\mathbb{P}\left[|\tilde{\mu}_{ZY} - \mu_{ZY}| > \frac{2\sigma_{ZY}}{\sqrt{m}}\right] \leq e^{-k/8}, \quad \mathbb{P}\left[|\tilde{\mu}_{ZX} - \mu_{ZX}| > \frac{2\sigma_{ZX}}{\sqrt{m}}\right] \leq e^{-k/8}.$$

**Step 2 (Union bound).** Define the events

$$A = \left\{|\tilde{\mu}_{ZY} - \mu_{ZY}| \leq \frac{2\sigma_{ZY}}{\sqrt{m}}\right\}, \quad B = \left\{|\tilde{\mu}_{ZX} - \mu_{ZX}| \leq \frac{2\sigma_{ZX}}{\sqrt{m}}\right\}.$$

By Step 1,  $\mathbb{P}[A^c] \leq e^{-k/8}$  and  $\mathbb{P}[B^c] \leq e^{-k/8}$ . Note that  $A$  and  $B$  are not independent (both are constructed from the same data), so we use the union bound:

$$\mathbb{P}[(A \cap B)^c] = \mathbb{P}[A^c \cup B^c] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq 2e^{-k/8}.$$

Setting  $k = \lceil 8 \ln(2/\delta) \rceil$  ensures  $2e^{-k/8} \leq \delta$ .

**Step 3 (Algebraic bound on the ratio).** On  $A \cap B$ , we decompose

$$\hat{\beta}_{\text{RM}} - \beta = \frac{\tilde{\mu}_{ZY}}{\tilde{\mu}_{ZX}} - \frac{\mu_{ZY}}{\mu_{ZX}} = \frac{\tilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{\mu}_{ZX}}{\tilde{\mu}_{ZX} \cdot \mu_{ZX}}.$$

Adding and subtracting  $\mu_{ZY}\mu_{ZX}$  in the numerator and using  $\mu_{ZY} = \beta\mu_{ZX}$ ,

$$\tilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{\mu}_{ZX} = \mu_{ZX}[(\tilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\tilde{\mu}_{ZX} - \mu_{ZX})].$$

The factor  $\mu_{ZX}$  cancels with the  $\mu_{ZX}$  in the denominator, giving

$$\hat{\beta}_{\text{RM}} - \beta = \frac{(\tilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\tilde{\mu}_{ZX} - \mu_{ZX})}{\tilde{\mu}_{ZX}}.$$

Taking absolute values and applying the triangle inequality to the numerator,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{|\tilde{\mu}_{ZY} - \mu_{ZY}| + |\beta| |\tilde{\mu}_{ZX} - \mu_{ZX}|}{|\tilde{\mu}_{ZX}|}.$$

**Step 4 (Substitute bounds).** On event  $A$ ,  $|\tilde{\mu}_{ZY} - \mu_{ZY}| \leq 2\sigma_{ZY}/\sqrt{m}$ . On event  $B$ ,  $|\tilde{\mu}_{ZX} - \mu_{ZX}| \leq 2\sigma_{ZX}/\sqrt{m}$ , and by the reverse triangle inequality,

$$|\tilde{\mu}_{ZX}| \geq |\mu_{ZX}| - |\tilde{\mu}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - \frac{2\sigma_{ZX}}{\sqrt{m}}.$$

The instrument strength condition (14) ensures  $2\sigma_{ZX}/\sqrt{m} < |\mu_{ZX}|$ , so the denominator is strictly positive. Substituting,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{2\sigma_{ZY}/\sqrt{m} + |\beta| \cdot 2\sigma_{ZX}/\sqrt{m}}{|\mu_{ZX}| - 2\sigma_{ZX}/\sqrt{m}} = \frac{2(\sigma_{ZY} + |\beta| \sigma_{ZX})}{|\mu_{ZX}| \sqrt{m} - 2\sigma_{ZX}}.$$

Since this holds on  $A \cap B$  and  $\mathbb{P}[(A \cap B)^c] \leq \delta$ , the result follows.  $\square$

## 5 The Median-of-Ratios (MoR) Estimator

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### Algorithm 3: Median-of-Ratios Estimator

---

**Input:** Sample  $(Y_i, X_i, Z_i)_{i=1}^n$ , number of blocks  $k$

**Output:** Estimate  $\hat{\beta}_{\text{MR}}$

Partition  $(Y_i, X_i, Z_i)_{i=1}^n$  into  $k$  blocks  $B_1, \dots, B_k$  of size  $m = \lfloor n/k \rfloor$ ;

**for**  $j = 1, \dots, k$  **do**

Compute the block means  $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$  and  $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$ ;  
 Compute block level IV estimates  $\hat{\beta}^{(j)} = \frac{\hat{\mu}_{ZY}^{(j)}}{\hat{\mu}_{ZX}^{(j)}}$ ;

**return**  $\hat{\beta}_{\text{MR}} = \text{median}(\hat{\beta}^{(1)}, \dots, \hat{\beta}^{(k)})$ ;

---

**Lemma 5.1** (Block-level error decomposition). *Under (1) and (A1), for each block  $j$ ,*

$$\hat{\beta}^{(j)} - \beta = \frac{\bar{\varepsilon}_j^Z}{\hat{\mu}_{ZX}^{(j)}}, \quad (16)$$

where  $\bar{\varepsilon}_j^Z = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i$  satisfies  $\mathbb{E}[\bar{\varepsilon}_j^Z] = 0$ .

*Proof.* Multiply  $Y_i = \beta X_i + \varepsilon_i$  by  $Z_i$  and average over block  $B_j$ :

$$\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i (\beta X_i + \varepsilon_i) = \beta \hat{\mu}_{ZX}^{(j)} + \bar{\varepsilon}_j^Z.$$

Dividing by  $\hat{\mu}_{ZX}^{(j)}$  gives  $\hat{\beta}^{(j)} = \beta + \bar{\varepsilon}_j^Z / \hat{\mu}_{ZX}^{(j)}$ . By (A1) and linearity of expectation,  $\mathbb{E}[\bar{\varepsilon}_j^Z] = \frac{1}{m} \sum_{i \in B_j} \mathbb{E}[Z_i \varepsilon_i] = 0$ .  $\square$

**Theorem 5.2** (MoR concentration). *Assume (A1)–(A3). Let  $\delta \in (0, 1)$ ,  $k = \lceil 8 \ln(1/\delta) \rceil$ , and  $m = \lfloor n/k \rfloor$ . If the instrument strength condition*

$$m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2} \quad (17)$$

*holds, then*

$$\mathbb{P} \left[ \left| \hat{\beta}_{\text{MR}} - \beta \right| > \frac{4\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{m}} \right] \leq \delta. \quad (18)$$

*Proof.* The proof applies the MoM counting argument directly to the block-level IV estimates  $\hat{\beta}^{(j)}$ .

**Step 1 (Two-event decomposition).** By Lemma 5.1,  $\left| \hat{\beta}^{(j)} - \beta \right| = \left| \bar{\varepsilon}_j^Z \right| / \left| \hat{\mu}_{ZX}^{(j)} \right|$ . To bound this ratio, we control the numerator above and the denominator below. For a denominator threshold  $D \in (0, |\mu_{ZX}|)$  and an error threshold  $t > 0$ , both to be chosen below, define

$$A_j(D) = \left\{ \left| \hat{\mu}_{ZX}^{(j)} \right| \geq D \right\}, \quad B_j(D, t) = \left\{ \left| \bar{\varepsilon}_j^Z \right| \leq tD \right\}.$$

On  $A_j(D) \cap B_j(D, t)$ , the denominator satisfies  $|\hat{\mu}_{ZX}^{(j)}| \geq D$  by definition of  $A_j(D)$ , so

$$|\hat{\beta}^{(j)} - \beta| = \frac{|\bar{\varepsilon}_j^Z|}{|\hat{\mu}_{ZX}^{(j)}|} \leq \frac{tD}{D} = t.$$

Hence  $\mathbb{P}\left[|\hat{\beta}^{(j)} - \beta| > t\right] \leq \mathbb{P}[A_j(D)^c] + \mathbb{P}[B_j(D, t)^c]$  by the union bound. For the remainder of this proof we set  $D = |\mu_{ZX}|/2$ , the symmetric choice that yields the cleanest constants.

**Step 2 (Chebyshev bounds).** The two failure events are the block-level analogues of those in the proof of Theorem 3.3, with  $\hat{\mu}_{ZX}$  replaced by the block mean  $\hat{\mu}_{ZX}^{(j)}$ ,  $\hat{\mu}_{Z\varepsilon}$  by  $\bar{\varepsilon}_j^Z$ , and the full sample size  $n$  by the block size  $m$ . The same reverse-triangle-inequality and Chebyshev arguments therefore give

$$\mathbb{P}[A_j^c] \leq \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}, \quad \mathbb{P}[B_j^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2}.$$

**Step 3 (Allocate the error budget and solve for  $t$ ).** Unlike in the standard IV proof, the target here is not  $\delta$  but the per-block failure probability  $1/4$ : this is the level at which the counting argument of Steps 4–5 boosts the median to the desired confidence. We require  $\mathbb{P}[A_j^c] + \mathbb{P}[B_j^c] \leq 1/4$  and allocate half the budget to each event.

*Budget for  $A_j^c$ .* Setting  $\mathbb{P}[A_j^c] \leq 1/8$  requires

$$\frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2} \leq \frac{1}{8} \iff m \geq \frac{32\sigma_{ZX}^2}{\mu_{ZX}^2},$$

which is exactly condition (17).

Setting  $\mathbb{P}[B_j^c] \leq 1/8$  and solving for  $t$ :

$$\frac{4\sigma_{Z\varepsilon}^2}{mt^2\mu_{ZX}^2} \leq \frac{1}{8} \implies t \geq \frac{\sqrt{32}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{m}}.$$

By the union bound, each block estimate satisfies

$$\mathbb{P}\left[|\hat{\beta}^{(j)} - \beta| > t\right] \leq \mathbb{P}[A_j^c] + \mathbb{P}[B_j^c] \leq \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.$$

**Step 4 (Counting argument).** Define  $\zeta_j = \mathbf{1}\{|\hat{\beta}^{(j)} - \beta| > t\}$ . By Step 3,  $\mathbb{E}[\zeta_j] \leq 1/4$ . Since the blocks are disjoint, the  $(\zeta_j)_{j=1}^k$  are independent. If  $|\hat{\beta}_{\text{MR}} - \beta| > t$ , the median of the block estimates lies outside  $(\beta - t, \beta + t)$ , which requires at least  $k/2$  blocks to satisfy  $|\hat{\beta}^{(j)} - \beta| > t$ . Therefore,

$$\mathbb{P}\left[|\hat{\beta}_{\text{MR}} - \beta| > t\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right].$$

**Step 5 (Hoeffding).** By Hoeffding's inequality, with overshoot  $k/2 - k/4 = k/4$ ,

$$\mathbb{P}\left[\sum_{j=1}^k \zeta_j \geq \frac{k}{2}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

Setting  $k = \lceil 8 \ln(1/\delta) \rceil$  gives  $e^{-k/8} \leq \delta$ , completing the proof.  $\square$

**Remark 5.3** (Logarithmic dependence on  $\delta$ ). *Through  $k = \lceil 8 \ln(1/\delta) \rceil$ , the bound (18) depends on  $1/\delta$  only logarithmically. This is precisely the improvement over the standard IV estimator anticipated in Remark 3.4: the median-of-block-estimates construction replaces the polynomial factor  $1/\sqrt{\delta}$  of (8) by  $\sqrt{\ln(1/\delta)}$ .*

**Remark 5.4** (The symmetric threshold is not optimal). *As in the standard IV estimator (Remark 3.5), the choice  $D = |\mu_{ZX}|/2$  was made for simplicity rather than optimality. Section 5.1 treats  $D$  as a free parameter.*

### 5.1 Optimised Threshold

The proof of Theorem 5.2 introduced a general threshold  $D$  but then fixed it symmetrically. Here we keep  $D$  free and choose it to minimise the resulting error bound, at the cost of a more involved Step 3.

**Theorem 5.5** (MoR concentration, optimised threshold). *Assume (A1)–(A3). Let  $\delta \in (0, 1)$ ,  $k = \lceil 8 \ln(1/\delta) \rceil$ , and  $m = \lfloor n/k \rfloor$ . Define*

$$\rho_{\text{MR}} = \frac{4\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

*If  $\rho_{\text{MR}} < 1$  (i.e.,  $m > 4\sigma_{ZX}^2/\mu_{ZX}^2$ ), then the choice  $D^* = (1 - \rho_{\text{MR}}^{1/3})|\mu_{ZX}|$  is optimal and*

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{MR}} - \beta\right| > \frac{2\sigma_{Z\epsilon}}{|\mu_{ZX}|\sqrt{m}(1 - \rho_{\text{MR}}^{1/3})^{3/2}}\right] \leq \delta. \quad (19)$$

*Proof.* Steps 1 and 2 are identical to those of Theorem 5.2 but with  $D$  left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A_j(D)^c] \leq \frac{\sigma_{ZX}^2/m}{(|\mu_{ZX}| - D)^2}, \quad \mathbb{P}[B_j(D, t)^c] \leq \frac{\sigma_{Z\epsilon}^2/m}{t^2 D^2}.$$

Requiring their sum to be at most  $\frac{1}{4}$  gives the joint constraint

$$\frac{\sigma_{ZX}^2}{m(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\epsilon}^2}{mt^2 D^2} \leq \frac{1}{4}. \quad (20)$$

**Step 3 (Optimise  $D$ , then solve for  $t$ ).** Writing  $D = \eta|\mu_{ZX}|$  with  $\eta \in (0, 1)$  and multiplying (20) through by 4, the constraint becomes  $\rho_{\text{MR}}/(1 - \eta)^2 + 4\sigma_{Z\epsilon}^2/(m\mu_{ZX}^2\eta^2 t^2) \leq 1$ . This is exactly the constraint optimised in the proof of Theorem 3.6, with  $\rho_{\text{IV}}$  replaced by  $\rho_{\text{MR}}$  and the

prefactor  $\sigma_{Z\varepsilon}^2/(\delta n \mu_{ZX}^2)$  by  $4\sigma_{Z\varepsilon}^2/(m \mu_{ZX}^2)$ . The same optimisation therefore yields  $\eta^* = 1 - \rho_{\text{MR}}^{1/3}$ , feasible if and only if  $\rho_{\text{MR}} < 1$ , and

$$t^* = \frac{2\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{m} (1 - \rho_{\text{MR}}^{1/3})^{3/2}}.$$

At this threshold each block satisfies  $\mathbb{P}[\left|\hat{\beta}^{(j)} - \beta\right| > t^*] \leq 1/4$ , and the counting argument and Hoeffding's inequality of Steps 4–5 in the proof of Theorem 5.2 then give  $\mathbb{P}[\left|\hat{\beta}_{\text{MR}} - \beta\right| > t^*] \leq \delta$ .  $\square$

## 5.2 Cantelli Improvement

As in the standard IV estimator (Section 3.2), the denominator failure event  $A_j(D)^c$  is one-sided: only a downward deviation of  $\hat{\mu}_{ZX}^{(j)}$  from  $\mu_{ZX}$  makes the denominator too small, while an upward deviation is harmless. Replacing the two-sided Chebyshev bound on this event by Cantelli's inequality (Lemma 3.8) sharpens the instrument strength condition.

**Theorem 5.6** (MoR concentration, Cantelli improvement). *Assume (A1)–(A3),  $\mu_{ZX} > 0$ , and let  $\delta \in (0, 1)$ ,  $k = \lceil 8 \ln(1/\delta) \rceil$ ,  $m = \lfloor n/k \rfloor$ . Write  $\gamma_{\text{MR}} = \sigma_{ZX}^2/(m \mu_{ZX}^2)$  and define*

$$h(\eta) = \frac{\eta^2((1 - \eta)^2 - 3\gamma_{\text{MR}})}{\gamma_{\text{MR}} + (1 - \eta)^2}, \quad \eta \in (0, 1). \quad (21)$$

If

$$m > \frac{3\sigma_{ZX}^2}{\mu_{ZX}^2}, \quad \text{equivalently,} \quad \gamma_{\text{MR}} < \frac{1}{3}, \quad (22)$$

then  $h(\eta) > 0$  on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{MR}} - \beta\right| > \frac{2\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{mh(\eta_{\text{MR}}^*)}}\right] \leq \delta, \quad (23)$$

where  $\eta_{\text{MR}}^*$  is the maximiser of  $h$  over its feasible domain.

*Proof.* We use the same two-event framework with  $D = \eta \mu_{ZX}$  kept general.

**Step 1 (Two-event decomposition).** Since  $\mu_{ZX} > 0$ , we drop the absolute value on the denominator and define  $A_j(D) = \{\hat{\mu}_{ZX}^{(j)} \geq D\}$  and  $B_j(D, t) = \{|\bar{\varepsilon}_j^Z| \leq tD\}$ . As in Theorem 5.2,  $|\hat{\beta}^{(j)} - \beta| \leq t$  on  $A_j(D) \cap B_j(D, t)$ .

**Step 2 (Cantelli on  $A_j^c$ , Chebyshev on  $B_j^c$ ).** The event  $A_j(D)^c$  is a one-sided downward deviation, so Lemma 3.8 applies with  $W = \hat{\mu}_{ZX}^{(j)}$ ,  $\sigma^2 = \sigma_{ZX}^2/m$ , and  $\tau = \mu_{ZX} - D$ . The event  $B_j(D, t)^c = \{|\bar{\varepsilon}_j^Z| > tD\}$  is two-sided, so Chebyshev remains appropriate. As in the proof of Theorem 3.9, these give

$$\mathbb{P}[A_j(D)^c] \leq \frac{\sigma_{ZX}^2/m}{\sigma_{ZX}^2/m + (\mu_{ZX} - D)^2}, \quad \mathbb{P}[B_j(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/m}{t^2 D^2}.$$

**Step 3 (Constraint and optimisation).** Setting the per-block failure probability to at most

1/4 and substituting  $D = \eta\mu_{ZX}$ ,  $\gamma_{\text{MR}} = \sigma_{Z\varepsilon}^2/(m\mu_{ZX}^2)$ :

$$\frac{\gamma_{\text{MR}}}{\gamma_{\text{MR}} + (1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{mt^2\eta^2\mu_{ZX}^2} \leq \frac{1}{4}.$$

Isolating the second term and solving for  $t^2$ :

$$t^2 \geq \frac{4\sigma_{Z\varepsilon}^2}{m\mu_{ZX}^2 h(\eta)},$$

where  $h(\eta)$  is defined in (21). The function  $h(\eta)$  is positive only when  $\frac{1}{4}(\gamma_{\text{MR}} + (1 - \eta)^2) > \gamma_{\text{MR}}$ , i.e.,  $(1 - \eta)^2 > 3\gamma_{\text{MR}}$ . The feasible region is non-empty if and only if  $\gamma_{\text{MR}} < \frac{1}{3}$ , which is condition (22). Minimising  $t$  over the feasible region by maximising  $h$  and applying Hoeffding and the union bound gives (23).  $\square$

**Remark 5.7** (No tractable form for  $\eta_{\text{MR}}^*$ ). *Exactly as for the standard IV estimator (Remark 3.11), the first-order condition  $h'(\eta) = 0$  reduces to a quartic in  $\eta$  with no tractable closed-form solution, in contrast to the clean form  $\eta^* = 1 - \rho_{\text{MR}}^{1/3}$  of the optimised Chebyshev version (Theorem 5.5). The bound (23) must therefore be evaluated numerically.*

## 6 Inference

### 6.1 Test Inversion

Fix a null hypothesis  $H_0: \beta = \beta_0$  and define

$$W_i(\beta_0) := Z_i(Y_i - \beta_0 X_i), \quad i = 1, \dots, n. \quad (24)$$

Under  $H_0$  we have  $W_i(\beta_0) = Z_i\varepsilon_i$ , so by (A1)  $\mathbb{E}[W_i(\beta_0)] = \mathbb{E}[Z_i\varepsilon_i] = 0$  and by (A3)  $\text{Var}(W_i(\beta_0)) = \sigma_{Z\varepsilon}^2 < \infty$ .

Dufour (1997) explains why test inversion is necessary under weak instruments. Let  $\mathcal{P}$  denote the family of joint distributions of  $(Y_i, X_i, Z_i)$  compatible with (A1) and (A3), note that (A2) is deliberately not imposed, and write  $\mu_{ZX}(P) := \mathbb{E}_P[Z_i X_i]$  for  $P \in \mathcal{P}$ . On the identification failure set

$$\mathcal{P}_0 := \{P \in \mathcal{P} : \mu_{ZX}(P) = 0\}$$

the parameter  $\beta$  is unidentified; exogeneity then forces  $\mu_{ZY}(P) = 0$  as well, so  $\mathbb{E}_P[Z_i(Y_i - bX_i)] = 0$  for every  $b \in \mathbb{R}$ . Since  $\mathcal{P}_0$  is non-empty, no confidence set procedure that is valid over all of  $\mathcal{P}$  can be bounded with probability one. A Wald-type interval built around a point estimate is therefore not generally valid, whereas inverting a test of  $H_0: \beta = \beta_0$  over a grid of null values remains so.

### 6.2 The Median-of-Means Anderson–Rubin Test

We partition the sample into  $k$  blocks as in the preceding sections and form, within each block  $B_j$ , the block mean of the moment (24). Because  $W_i(\beta_0) = Z_i Y_i - \beta_0 Z_i X_i$  is linear in  $\beta_0$ , the

block mean is an *affine* function of  $\beta_0$ :

$$\bar{W}_j(\beta_0) := \frac{1}{m} \sum_{i \in B_j} W_i(\beta_0) = \hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)}, \quad j = 1, \dots, k. \quad (25)$$

The median-of-means AR statistic is the coordinate-wise median of these block means,

$$\widetilde{W}(\beta_0) := \text{med}(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)). \quad (26)$$

---

**Algorithm 4:** Median-of-Means Anderson–Rubin Test

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**Input:** Sample  $(Y_i, X_i, Z_i)_{i=1}^n$ , number of blocks  $k$ , null value  $\beta_0$ , threshold  $\tau_n(\delta)$

**Output:** Decision: reject or fail to reject  $H_0: \beta = \beta_0$

Partition  $(Y_i, X_i, Z_i)_{i=1}^n$  into  $k$  blocks  $B_1, \dots, B_k$  of size  $m = \lfloor n/k \rfloor$ ;

**for**  $j = 1, \dots, k$  **do**

    Compute  $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$  and  $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$ ;  
    Form  $\bar{W}_j(\beta_0) = \hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)}$ ;

$\widetilde{W}(\beta_0) \leftarrow \text{med}(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0))$ ;

**return** reject if  $|\widetilde{W}(\beta_0)| > \tau_n(\delta)$ , else fail to reject;

---

**Theorem 6.1** (Size of the MoM-AR test). *Assume (A1)–(A3). Fix  $\delta \in (0, 1)$ , set  $k = \lceil 8 \ln(1/\delta) \rceil$  and  $m = \lfloor n/k \rfloor$ , and define*

$$\tau_n(\delta) := \sigma_{Z\varepsilon} \sqrt{\frac{32 \ln(1/\delta)}{n}}. \quad (27)$$

Then under  $H_0: \beta = \beta_0$ ,

$$\mathbb{P}_{H_0} \left[ |\widetilde{W}(\beta_0)| > \tau_n(\delta) \right] \leq \delta. \quad (28)$$

Consequently the test of Algorithm 4 has size at most  $\delta$ .

*Proof.* The proof reduces the statistic to the scalar median-of-means estimator of Section 2.3 and applies its concentration bound.

**Step 1 (Moments of the block moment under  $H_0$ ).** Under  $H_0: \beta = \beta_0$  we have  $W_i(\beta_0) = Z_i \varepsilon_i$ , so the  $(W_i(\beta_0))_{i=1}^n$  are i.i.d. with  $\mathbb{E}[W_i(\beta_0)] = \mathbb{E}[Z_i \varepsilon_i] = 0$  by (A1) and  $\text{Var}(W_i(\beta_0)) = \sigma_{Z\varepsilon}^2 < \infty$  by (A3).

**Step 2 (Reduction to the scalar MoM estimator).** By (25) and (26),  $\widetilde{W}(\beta_0)$  is the coordinate-wise median of the  $k$  block means of  $(W_i(\beta_0))_{i=1}^n$ ; that is, it is exactly the scalar median-of-means estimator of Algorithm 1 applied to this sequence, estimating its mean  $\mu = 0$ . Theorem 2.4, taken with  $\sigma^2 = \sigma_{Z\varepsilon}^2$  and  $k = \lceil 8 \ln(1/\delta) \rceil$ , therefore gives

$$\mathbb{P}_{H_0} \left[ |\widetilde{W}(\beta_0)| > \sigma_{Z\varepsilon} \sqrt{\frac{32 \ln(1/\delta)}{n}} \right] \leq \delta.$$

The deviation on the left is precisely  $\tau_n(\delta)$  of (27), so this is (28).  $\square$

### 6.3 The MoM-AR Confidence Set

Inverting the test of Theorem 6.1 produces a confidence set.

**Definition 6.2** (MoM-AR confidence set). *For  $\delta \in (0, 1)$  and  $\tau_n(\delta)$  as in (27), the MoM-AR confidence set is*

$$CS_{1-\delta} := \{\beta_0 \in \mathbb{R} : |\widetilde{W}(\beta_0)| \leq \tau_n(\delta)\}. \quad (29)$$

**Theorem 6.3** (Coverage). *Assume (A1)–(A3) and let  $k = \lceil 8 \ln(1/\delta) \rceil$ . Then*

$$\mathbb{P}_\beta[\beta \in CS_{1-\delta}] \geq 1 - \delta.$$

*Proof.* The result is the test-inversion dual of the size bound. By Definition 6.2, the true parameter lies in the confidence set exactly when the statistic evaluated at the truth stays below the threshold:

$$\beta \in CS_{1-\delta} \iff |\widetilde{W}(\beta)| \leq \tau_n(\delta).$$

Evaluating the moment at the true parameter gives  $W_i(\beta) = Z_i \varepsilon_i$ , which has mean zero by (A1). Applying the size bound of Theorem 6.1 at the true null  $\beta_0 = \beta$  therefore yields  $\mathbb{P}_\beta[|\widetilde{W}(\beta)| > \tau_n(\delta)] \leq \delta$ . Taking complements,

$$\mathbb{P}_\beta[\beta \in CS_{1-\delta}] = \mathbb{P}_\beta[|\widetilde{W}(\beta)| \leq \tau_n(\delta)] \geq 1 - \delta. \quad \square$$

### 6.4 Geometry of the Confidence Set

**Theorem 6.4** (Piecewise-affine structure). *The map  $\beta_0 \mapsto \widetilde{W}(\beta_0)$  is continuous and piecewise affine. There exist finitely many breakpoints  $\beta^{(1)} < \beta^{(2)} < \dots < \beta^{(M)}$ , with  $M \leq \binom{k}{2}$ , such that on each open interval of  $\mathbb{R} \setminus \{\beta^{(1)}, \dots, \beta^{(M)}\}$  the function  $\widetilde{W}$  coincides with a single block mean  $\bar{W}_j$  and is therefore affine.*

*Proof.* The median of  $k$  numbers is the value occupying rank  $r := \lceil k/2 \rceil$  once the numbers are sorted. We therefore study how the sorted order of  $\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)$  varies with  $\beta_0$ .

**Step 1 (Crossings are finite).** For a pair  $i \neq j$ , the equation  $\bar{W}_i(\beta_0) = \bar{W}_j(\beta_0)$  reads  $\hat{\mu}_{ZY}^{(i)} - \beta_0 \hat{\mu}_{ZX}^{(i)} = \hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)}$ , i.e.  $(\hat{\mu}_{ZX}^{(j)} - \hat{\mu}_{ZX}^{(i)})\beta_0 = \hat{\mu}_{ZY}^{(j)} - \hat{\mu}_{ZY}^{(i)}$ . If  $\hat{\mu}_{ZX}^{(i)} \neq \hat{\mu}_{ZX}^{(j)}$  this has a unique solution; if the slopes are equal the lines are parallel and never cross (or coincide, in which case the pair may be merged). Hence the crossing set  $\mathcal{C} := \{\beta_0 : \bar{W}_i(\beta_0) = \bar{W}_j(\beta_0) \text{ for some } i \neq j\}$  satisfies  $|\mathcal{C}| \leq \binom{k}{2}$ . Enumerate it as  $\beta^{(1)} < \dots < \beta^{(M)}$  and set  $\beta^{(0)} := -\infty$ ,  $\beta^{(M+1)} := +\infty$ .

**Step 2 (The order is constant between crossings).** Fix an interval  $I_\ell := (\beta^{(\ell)}, \beta^{(\ell+1)})$ . For any pair  $i \neq j$  the difference  $D_{ij}(\beta_0) := \bar{W}_i(\beta_0) - \bar{W}_j(\beta_0)$  is affine, hence continuous, and vanishes only on  $\mathcal{C}$ . Since  $I_\ell \cap \mathcal{C} = \emptyset$ ,  $D_{ij}$  has constant sign on  $I_\ell$ . As this holds for every pair, the total ordering of  $\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)$  is constant on  $I_\ell$ ; let  $\pi_\ell$  be the permutation realising it, so that  $\bar{W}_{\pi_\ell(1)}(\beta_0) \leq \dots \leq \bar{W}_{\pi_\ell(k)}(\beta_0)$  throughout  $I_\ell$ .

**Step 3 (The median is one block mean on each interval).** By Step 2 the rank- $r$  value on  $I_\ell$  is always supplied by the same index, so  $\widetilde{W}(\beta_0) = \bar{W}_{\pi_\ell(r)}(\beta_0)$  for all  $\beta_0 \in I_\ell$ ; this is affine. With at most  $M + 1 \leq \binom{k}{2} + 1$  intervals,  $\widetilde{W}$  is piecewise affine.

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**Step 4 (Continuity).** Let  $i = \pi_{\ell-1}(r)$  and  $j = \pi_\ell(r)$  be the active median indices on either side of a breakpoint  $\beta^{(\ell)}$ . If  $i = j$  there is nothing to prove. If  $i \neq j$ , the rank- $r$  slot changes occupant across  $\beta^{(\ell)}$ , which can occur only if lines  $i$  and  $j$  exchange ranks there; by continuity of  $D_{ij}$  this requires  $\bar{W}_i(\beta^{(\ell)}) = \bar{W}_j(\beta^{(\ell)})$ . Hence the left and right limits of  $\widetilde{W}$  at  $\beta^{(\ell)}$  coincide with this common value, and  $\widetilde{W}$  is continuous. (If three or more lines meet at  $\beta^{(\ell)}$ , any reshuffling is confined to the indices sharing the common value there, and the same conclusion holds.)  $\square$

**Corollary 6.5** (Candidate boundary points). *Every boundary point of  $CS_{1-\delta}$  lies in the set*

$$\mathcal{B}_{\text{cand}} := \bigcup_{j: \hat{\mu}_{ZX}^{(j)} \neq 0} \left\{ \frac{\hat{\mu}_{ZY}^{(j)} - \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}}, \frac{\hat{\mu}_{ZY}^{(j)} + \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}} \right\}, \quad (30)$$

*a set of at most  $2k$  explicitly computable points.*

*Proof.* A boundary point  $\beta_0$  of  $CS_{1-\delta}$  lies on the boundary of the band, so  $|\widetilde{W}(\beta_0)| = \tau_n(\delta)$ , i.e.  $\widetilde{W}(\beta_0) = +\tau_n(\delta)$  or  $\widetilde{W}(\beta_0) = -\tau_n(\delta)$ . By Theorem 6.4,  $\beta_0$  lies in the closure of an interval on which  $\widetilde{W}$  coincides with a single block mean  $\bar{W}_j$ , so it solves  $\bar{W}_j(\beta_0) = \pm\tau_n(\delta)$  for some  $j$ . Two cases arise, according to the slope  $-\hat{\mu}_{ZX}^{(j)}$  of that piece.

If  $\hat{\mu}_{ZX}^{(j)} \neq 0$ , then  $\bar{W}_j$  is strictly monotone and  $\hat{\mu}_{ZY}^{(j)} - \beta_0 \hat{\mu}_{ZX}^{(j)} = \pm\tau_n(\delta)$  has the unique solutions

$$\beta_0 = \frac{\hat{\mu}_{ZY}^{(j)} \mp \tau_n(\delta)}{\hat{\mu}_{ZX}^{(j)}},$$

namely the two values contributed by block  $j$  in (30). If  $\hat{\mu}_{ZX}^{(j)} = 0$ , then  $\bar{W}_j \equiv \hat{\mu}_{ZY}^{(j)}$  is constant and equals  $\pm\tau_n(\delta)$  only in the non-generic event  $\hat{\mu}_{ZY}^{(j)} = \pm\tau_n(\delta)$ ; such a block contributes no isolated boundary point. Ranging over the  $k$  blocks, each of which contributes at most two candidates, gives  $|\mathcal{B}_{\text{cand}}| \leq 2k$ .  $\square$

**Corollary 6.6** (Finite union of intervals).  *$CS_{1-\delta}$  is a finite union of at most  $k + 1$  closed intervals, each possibly a single point or unbounded.*

*Proof.* By Theorem 6.4,  $|\widetilde{W}|$  is continuous and piecewise affine. The set  $CS_{1-\delta} = \{\beta_0 : |\widetilde{W}(\beta_0)| \leq \tau_n(\delta)\}$  is the preimage of the closed interval  $[-\tau_n(\delta), \tau_n(\delta)]$  under this continuous function, hence closed; being a sublevel set of a piecewise-affine function with finitely many pieces, it is a finite union of closed intervals.

To count the components, note that the endpoints of these intervals are boundary points of  $CS_{1-\delta}$ , and by Corollary 6.5 there are at most  $2k$  such points. Ordering them partitions  $\mathbb{R}$  into at most  $2k + 1$  open intervals, on each of which  $|\widetilde{W}| - \tau_n(\delta)$  has constant sign and so lies either entirely inside or entirely outside the band. Since the in-band and out-of-band intervals alternate along the line, at most  $\lceil (2k + 1)/2 \rceil = k + 1$  of them lie in the band. Merging any in-band intervals that meet at a shared boundary point can only reduce this count, so  $CS_{1-\delta}$  has at most  $k + 1$  components.  $\square$

## 6.5 Monotonicity and Single-Interval Confidence Sets

The slope of  $\widetilde{W}$  on any affine piece is  $-\hat{\mu}_{ZX}^{(j)}$  for the active index  $j$ . When all block slopes share a sign,  $\widetilde{W}$  is monotonic and the confidence set is a single interval.

**Proposition 6.7** (Deterministic single-interval condition). *If the block sums  $\hat{\mu}_{ZX}^{(1)}, \dots, \hat{\mu}_{ZX}^{(k)}$  all share the same sign, then  $\widetilde{W}$  is monotonic on  $\mathbb{R}$  and  $CS_{1-\delta}$  is a single (possibly empty or unbounded) interval.*

*Proof.* By Theorem 6.4 the slope of  $\widetilde{W}$  on each affine piece equals  $-\hat{\mu}_{ZX}^{(j)}$  for the active index  $j$ . If all  $\hat{\mu}_{ZX}^{(j)} > 0$  then every such slope is negative, so the continuous piecewise-affine function  $\widetilde{W}$  is non-increasing on  $\mathbb{R}$ ; if all  $\hat{\mu}_{ZX}^{(j)} < 0$  it is non-decreasing. In either case  $\widetilde{W}$  is monotone, so each of the level sets  $\{\widetilde{W} \leq \tau_n(\delta)\}$  and  $\{\widetilde{W} \geq -\tau_n(\delta)\}$  is a half-line (a ray). The confidence set  $CS_{1-\delta}$  is their intersection, hence a single interval, which may be empty or unbounded.  $\square$

**Proposition 6.8** (Single-interval condition, Chebyshev). *Assume (A1)–(A3) with  $\mu_{ZX} \neq 0$ . If*

$$m \geq \frac{k \sigma_{ZX}^2}{\delta \mu_{ZX}^2}, \quad (31)$$

*then with probability at least  $1 - \delta$  all block sums  $\hat{\mu}_{ZX}^{(j)}$  share the sign of  $\mu_{ZX}$ ; consequently  $\widetilde{W}$  is monotone and  $CS_{1-\delta}$  is a single interval.*

*Proof.* Assume without loss of generality that  $\mu_{ZX} > 0$ ; the case  $\mu_{ZX} < 0$  is symmetric. By Proposition 6.7 it suffices to show that all block sums are positive with probability at least  $1 - \delta$ , since sign-consistency then forces  $\widetilde{W}$  to be monotone. Sign-consistency fails on the event

$$\mathcal{F} = \bigcup_{j=1}^k \{\hat{\mu}_{ZX}^{(j)} \leq 0\}.$$

**Step 1 (Chebyshev on each block).** A non-positive block sum is a large downward deviation from  $\mu_{ZX} > 0$ : if  $\hat{\mu}_{ZX}^{(j)} \leq 0$  then  $|\hat{\mu}_{ZX}^{(j)} - \mu_{ZX}| = \mu_{ZX} - \hat{\mu}_{ZX}^{(j)} \geq \mu_{ZX}$ , so  $\{\hat{\mu}_{ZX}^{(j)} \leq 0\} \subseteq \{|\hat{\mu}_{ZX}^{(j)} - \mu_{ZX}| \geq \mu_{ZX}\}$ . Since  $\text{Var}(\hat{\mu}_{ZX}^{(j)}) = \sigma_{ZX}^2/m$ , Chebyshev's inequality applied with threshold  $\mu_{ZX}$  gives

$$\mathbb{P}[\hat{\mu}_{ZX}^{(j)} \leq 0] \leq \frac{\sigma_{ZX}^2/m}{\mu_{ZX}^2} = \frac{\sigma_{ZX}^2}{m\mu_{ZX}^2}.$$

**Step 2 (Union bound over blocks).** Summing over the  $k$  blocks,

$$\mathbb{P}[\mathcal{F}] \leq \sum_{j=1}^k \mathbb{P}[\hat{\mu}_{ZX}^{(j)} \leq 0] \leq \frac{k\sigma_{ZX}^2}{m\mu_{ZX}^2},$$

which is at most  $\delta$  precisely when  $m \geq k\sigma_{ZX}^2/(\delta\mu_{ZX}^2)$ , i.e. condition (31). On the complement  $\mathcal{F}^c$ , all block sums share the sign of  $\mu_{ZX}$ , and Proposition 6.7 makes  $CS_{1-\delta}$  a single interval.  $\square$

**Remark 6.9** (Cantelli refinement). *Replacing Chebyshev's inequality by the one-sided Cantelli bound (Lemma 3.8) in Step 1 sharpens the per-block probability, and hence the constant in*

condition (31). As for the standard IV estimator (Remark 3.10), the resulting improvement is minor, so we retain the Chebyshev form.

## 6.6 Computation

The confidence set can be computed exactly and in closed form. The idea is to enumerate the crossing points of the block-mean lines  $\bar{W}_1, \dots, \bar{W}_k$  to obtain the affine pieces of  $\widetilde{W}$  (Theorem 6.4), identify the active median line on each piece, and retain the sub-intervals on which this line lies within the band  $[-\tau_n(\delta), \tau_n(\delta)]$ . Algorithm 5 collects these steps.

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### Algorithm 5: MoM-AR Confidence Set

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**Input:** Sample  $(Y_i, X_i, Z_i)_{i=1}^n$ , number of blocks  $k$ , threshold  $\tau_n(\delta)$   
**Output:** Confidence set  $CS_{1-\delta}$   
 Partition  $(Y_i, X_i, Z_i)_{i=1}^n$  into  $k$  blocks  $B_1, \dots, B_k$  of size  $m = \lfloor n/k \rfloor$ ;  
**for**  $j = 1, \dots, k$  **do**  
   Compute  $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$  and  $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$ ;  
 $\mathcal{C} \leftarrow \emptyset$ ;  
**foreach** pair  $1 \leq i < j \leq k$  with  $\hat{\mu}_{ZX}^{(i)} \neq \hat{\mu}_{ZX}^{(j)}$  **do**  
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{(\hat{\mu}_{ZY}^{(j)} - \hat{\mu}_{ZY}^{(i)}) / (\hat{\mu}_{ZX}^{(j)} - \hat{\mu}_{ZX}^{(i)})\}$ ;  
 Sort  $\mathcal{C}$  as  $\beta^{(1)} < \dots < \beta^{(M)}$  and set  $\beta^{(0)} = -\infty, \beta^{(M+1)} = +\infty$ ;  
 $CS_{1-\delta} \leftarrow \emptyset$ ;  
**for**  $\ell = 0, \dots, M$  **do**  
   Choose any  $\beta_\star \in (\beta^{(\ell)}, \beta^{(\ell+1)})$ ;  
   Find the active median index  $a$  with  $\bar{W}_a(\beta_\star) = \text{med}(\bar{W}_1(\beta_\star), \dots, \bar{W}_k(\beta_\star))$ ;  
   **if**  $\hat{\mu}_{ZX}^{(a)} \neq 0$  **then**  
      $J \leftarrow$  closed interval with endpoints  $(\hat{\mu}_{ZY}^{(a)} \pm \tau_n(\delta)) / \hat{\mu}_{ZX}^{(a)}$ ;  
   **else**  
      $J \leftarrow \mathbb{R}$  if  $|\hat{\mu}_{ZY}^{(a)}| \leq \tau_n(\delta)$ , else  $J \leftarrow \emptyset$ ;  
    $CS_{1-\delta} \leftarrow CS_{1-\delta} \cup ([\beta^{(\ell)}, \beta^{(\ell+1)}] \cap J)$ ;  
**return**  $CS_{1-\delta}$ , merging intervals that share an endpoint;

---

On each interval  $(\beta^{(\ell)}, \beta^{(\ell+1)})$  the statistic  $\widetilde{W}$  coincides with a single affine piece  $\bar{W}_a$ , so intersecting the band constraint  $|\bar{W}_a(\beta_0)| \leq \tau_n(\delta)$  with that interval is exact. By Corollary 6.6 the returned set consists of at most  $k + 1$  closed intervals, whose endpoints lie among the  $2k$  candidates of Corollary 6.5.

**Remark 6.10** (Feasibility and choice of variance estimator). *In practice,  $\sigma_{Z\varepsilon}$  is unknown and has to be estimated with a consistent estimator. A feasible test would replace this by a consistent estimator  $\hat{\sigma}_{Z\varepsilon}$ . This estimator must be computed independently of the null value  $\beta_0$ ; test inversion compares the varying statistic  $\widetilde{W}(\beta_0)$  against a single band across a grid of  $\beta_0$  values. Forming residuals at the tested value is inconsistent whenever  $\beta_0 \neq \beta$  since  $Y_i - \beta_0 X_i = \varepsilon_i + (\beta - \beta_0) X_i$ ,*

$$\frac{1}{n} \sum_{i=1}^n (Z_i(Y_i - \beta_0 X_i))^2 \xrightarrow{p} \sigma_{Z\varepsilon}^2 + 2(\beta - \beta_0) \mathbb{E}[Z^2 X \varepsilon] + (\beta - \beta_0)^2 \mathbb{E}[(ZX)^2],$$

which grows as  $\beta_0$  departs from  $\beta$ . This would lead to the band widening with the signal it is

trying to detect, leading to a loss of power and a collapse of the constant-threshold geometry of Section 6.4.

A  $\beta_0$ -free estimate could be obtained instead from residuals at a preliminary consistent estimator. However, under the heavy tails permitted by (A3),  $Z_\varepsilon$  may have no finite fourth moment, in which case the estimator  $\frac{1}{n} \sum_i (Z_i \hat{\varepsilon}_i)^2$  has infinite variance; the central limit theorem then no longer applies, so it converges slowly and behaves erratically. A robust scale estimator is therefore preferable.

The self-normalised variant discussed in Section 6.7 avoids the matter altogether, cancelling  $\sigma_{Z_\varepsilon}$  rather than estimating it.

## 6.7 Self-Normalised Anderson-Rubin Test

**Definition 6.11** (Self-normalised statistic). For a vector  $v = (v_1, \dots, v_k) \in \mathbb{R}^k$ , let  $\text{MAD}(v) = \text{med}_j |v_j - \text{med}_l v_l|$  denote its median absolute deviation. With  $\bar{W}_j$  and  $\widetilde{W}$  as in (25)–(26), the self-normalised statistic at  $\beta_0$  is

$$T(\beta_0) = \frac{|\widetilde{W}(\beta_0)|}{\hat{s}(\beta_0)}, \quad \hat{s}(\beta_0) = \text{MAD}(\bar{W}_1(\beta_0), \dots, \bar{W}_k(\beta_0)).$$

Given a critical value  $c_{k,\delta}$ , the test rejects  $H_0: \beta = \beta_0$  when  $T(\beta_0) > c_{k,\delta}$ .

**Proposition 6.12** (Scale invariance). Under  $H_0$ , the distribution of  $T(\beta_0)$  does not depend on  $\sigma_{Z_\varepsilon}$ . The same holds for any scale functional  $\hat{s}$  that is homogeneous of degree one.

*Proof.* Under  $H_0$ ,  $\bar{W}_j(\beta_0) = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i = \sigma_{Z_\varepsilon} G_j$ , where  $G_j = \frac{1}{m} \sum_{i \in B_j} Z_i \varepsilon_i / \sigma_{Z_\varepsilon}$ . The summands  $Z_i \varepsilon_i / \sigma_{Z_\varepsilon}$  have mean 0 and variance 1 by (A1) and (A3), so the joint distribution  $(G_1, \dots, G_k)$  are free of  $\sigma_{Z_\varepsilon}$ . Since med and MAD are homogeneous of degree one,

$$T(\beta_0) = \frac{|\text{med}_j \sigma_{Z_\varepsilon} G_j|}{\text{MAD}_j(\sigma_{Z_\varepsilon} G_j)} = \frac{\sigma_{Z_\varepsilon} |\text{med}_j G_j|}{\sigma_{Z_\varepsilon} \text{MAD}_j(G_j)} = \frac{|\text{med}_j G_j|}{\text{MAD}_j(G_j)},$$

and a function of  $(G_1, \dots, G_k)$  alone; its distribution therefore does not involve  $\sigma_{Z_\varepsilon}$ . The argument uses only  $\text{med}(\sigma v) = \sigma \text{med}(v)$  and  $\hat{s}(\sigma v) = \sigma \hat{s}(v)$ , so it holds for any degree-one homogeneous  $\hat{s}$ .  $\square$

**Proposition 6.13** (Asymptotic null distribution). Assume (A1)–(A3), with  $k$  fixed and  $m \rightarrow \infty$ . Under  $H_0$ ,

$$T(\beta_0) \implies R_k := \frac{|\text{med}(\xi_1, \dots, \xi_k)|}{\text{MAD}(\xi_1, \dots, \xi_k)}, \quad \xi_1, \dots, \xi_k \stackrel{iid}{\sim} N(0, 1),$$

and the distribution of  $R_k$  depends only on  $k$ .

*Proof.* Fix a block. By (A1) and (A3) the standardised summands  $Z_i \varepsilon_i / \sigma_{Z_\varepsilon}$  are i.i.d. with mean 0 and variance 1, so the central limit theorem gives  $\sqrt{m} G_j = \frac{1}{\sqrt{m}} \sum_{i \in B_j} Z_i \varepsilon_i / \sigma_{Z_\varepsilon} \Rightarrow \xi_j \sim N(0, 1)$ . The blocks use disjoint observations, so  $(\sqrt{m} G_1, \dots, \sqrt{m} G_k) \Rightarrow (\xi_1, \dots, \xi_k)$  with independent coordinates. Writing  $T(\beta_0) = |\text{med}_j \sqrt{m} G_j| / \text{MAD}_j(\sqrt{m} G_j)$ , the factor  $\sqrt{m}$  cancels between numerator and denominator; as med and MAD are continuous, the continuous mapping theorem

yields  $T(\beta_0) \Rightarrow R_k$ . The limit is a fixed functional of  $k$  standard normals, so its distribution depends only on  $k$ .  $\square$

The critical value  $c_{k,\delta}$  is the  $(1-\delta)$  quantile of  $R_k$ ; since  $R_k$  depends only on  $k$ , it is computed once by simulation and tabulated in Section 9.1

**Corollary 6.14** (Confidence set). *Inverting the self-normalised test gives*

$$CS_{1-\delta} = \{\beta_0 \in \mathbb{R} : |\widetilde{W}(\beta_0)| \leq c_{k,\delta} \hat{s}(\beta_0)\},$$

*a finite union of closed intervals.*

*Proof.* By Theorem 6.4,  $\widetilde{W}$  is continuous and piecewise affine;  $\hat{s}(\beta_0) = \text{MAD}_j(\bar{W}_j(\beta_0))$  is a median of absolute values of affine functions, hence also continuous and piecewise affine. On any interval where both coincide with single affine pieces  $a(\beta_0)$  and  $b(\beta_0) \geq 0$ , the defining condition  $-c_{k,\delta} b(\beta_0) \leq a(\beta_0) \leq c_{k,\delta} b(\beta_0)$  is a pair of linear inequalities, hence an interval. As there are finitely many pieces,  $CS_{1-\delta}$  is a finite union of closed intervals.  $\square$

**Remark 6.15** (An exactly distribution-free variant). *If  $Z\varepsilon$  is symmetric about 0 under  $H_0$ , the signs of the block means are independent fair coin flips, so  $N_+(\beta_0) = \#\{j : \bar{W}_j(\beta_0) > 0\} \sim \text{Binomial}(k, \frac{1}{2})$  exactly, free of both  $\sigma_{Z\varepsilon}$  and the tail shape. Rejecting when  $N_+(\beta_0)$  is far from  $k/2$  gives a finite-sample-exact, distribution-free test, at the cost of the symmetry assumption and of lower power, since it uses only the signs of the block means.*

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## 7 Simulation

### 7.1 Data Generation

Data is simulated from

$$Y = \beta X + \varepsilon_Y$$

where  $X$  is generated by

$$X = \mu_{ZX} Z + \varepsilon_X$$

with  $Z \sim N(0, 1)$ . In order to test the models under different regimes, we utilise three different shock families for  $\varepsilon_Y$ : Normal ( $\sim N(0, 1)$ ), t ( $\sim t(\nu)$ ), and centered Pareto ( $\sim \text{Pareto}(\alpha)$ ). The base values are:  $\beta = 1, \mu_{ZX} = 1, \sigma_{ZX}^2 = 2.5, \sigma_{Z\varepsilon}^2 = 1, \rho = 0.5, \delta = 0.05$ , and  $n = 2000$  unless stated otherwise. At this n, all strong instrument conditions hold.

### 7.2 Competitor Estimator

Besides the standard IV estimator of Section 3, we include Catoni's M-estimator of the mean (Catoni, 2012) as a competitor. Alongside Median-of-Means, it is the other canonical sub-Gaussian mean estimator under the sole assumption of finite variance (Devroye et al., 2016), which makes it the natural robust benchmark for the estimators of Sections 4–5.

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For a sample  $(V_i)_{i=1}^n$  with mean  $\mu_V$  and variance  $\sigma_V^2$ , Catoni's estimator  $\hat{\mu}_V^{\text{Cat}}$  is the root of

$$\sum_{i=1}^n \psi(\alpha(V_i - \theta)) = 0, \quad \psi(u) = \text{sign}(u) \ln\left(1 + |u| + \frac{u^2}{2}\right),$$

where the bounded influence function  $\psi$  replaces the identity implicit in the sample mean, so that no single observation can move the estimate by more than  $O(1/\alpha)$ . The tuning parameter  $\alpha$  is chosen from the target confidence level and the noise scale as

$$\alpha = \sqrt{\frac{2 \ln(2/\delta)}{nv \left(1 + \frac{2 \ln(2/\delta)}{n-2 \ln(2/\delta)}\right)}},$$

which, for any variance bound  $v \geq \sigma_V^2$ , yields a sub-Gaussian deviation guarantee at confidence level  $1 - \delta$  (Catoni, 2012). In the simulations  $v$  is not treated as known: it is replaced by a preliminary Median-of-Means estimate of the variance, so that the tuning itself remains valid under heavy tails and the comparison does not favour Catoni through oracle knowledge of scale.

The IV analogue mirrors the Ratio-of-Medians construction: both cross-moments are estimated by Catoni's method and the ratio is taken,

$$\hat{\beta}_{\text{Cat}} = \frac{\hat{\mu}_{ZY}^{\text{Cat}}}{\hat{\mu}_{ZX}^{\text{Cat}}},$$

with the confidence budget split evenly across the two coordinates, i.e. each estimate is tuned at level  $\delta/2$ , matching the allocation used by the Ratio-of-Medians estimator in Theorem 4.1. Since the root equation is strictly decreasing in  $\theta$ , each estimate is computed exactly by bisection.

## 7.3 Simulation Results

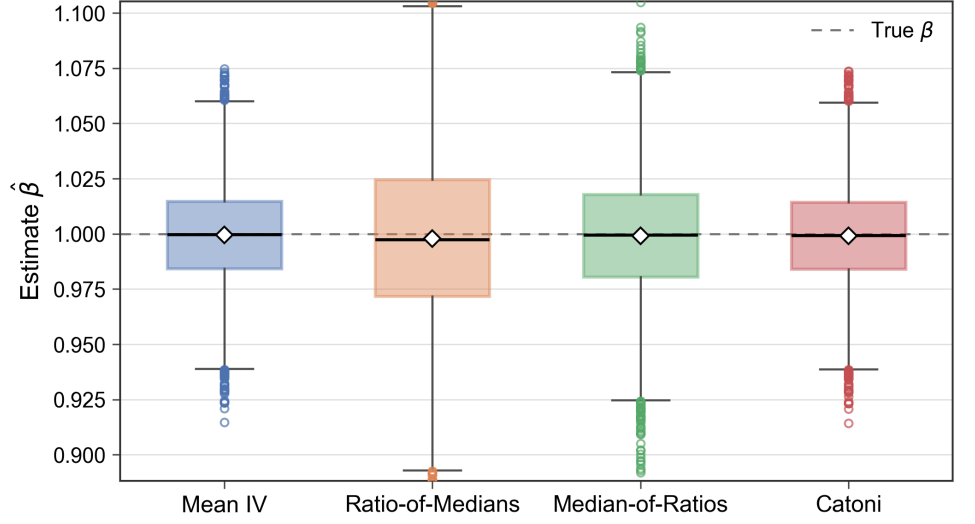
### 7.3.1 Cost of Robustness under Gaussian Errors

As can be seen in Figure 1, all four estimators are unbiased, but their spreads differ. Mean IV and Catonig both have a standard deviation of 0.0224, MoR of 0.0274, and RoM 0.0396. These results are exactly as expected; under light tails, the sample mean is already the efficient estimator, so blocking and taking medians only throws information away. Catoni has equal results to the Mean IV because its influence function is nearly linear on well-behaved data. That is, the Catoni estimator is the mean until extreme observations appear.

### 7.3.2 Benefit of Robustness under heavy tails

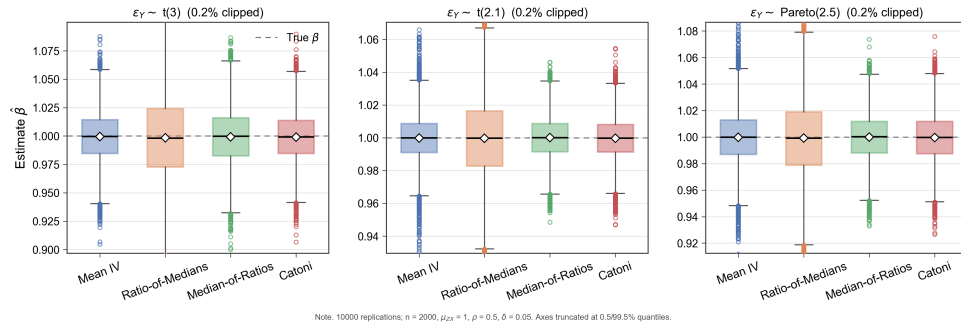
As per Figure 2, the benefit of MoM-type estimators becomes clear under  $T(2.1)$  errors; the RMSE is 0.0128 for MoR and 0.0130 for Catoni against 0.0189 for Mean IV. The error quantile is 0.0341 under MoR and 0.0446 under Mean IV. RoM is clearly insufficient in all cases.

The IQR of the Mean IV and MoR estimators are very similar; the difference between the two is in the outliers, with MoR being much less susceptible to extreme values. This is expected, as MoM-type estimators do not promise variance bounds, but tail bounds, so the benefit lives in the tails. The Pareto panel shows no bias for any of the estimators, showing that all estimators



Note. 10000 replications;  $n = 2000$ , Gaussian errors,  $\mu_{ZX} = 1$ ,  $\rho = 0.5$ ,  $\delta = 0.05$ . Axis truncated at 0.5/99.5% quantiles (0.16% of points outside)

Figure 1: Boxplot showing the cost of robustness under Gaussian errors



Note. 10000 replications;  $n = 2000$ ,  $\mu_{ZX} = 1$ ,  $\rho = 0.5$ ,  $\delta = 0.05$ . Axes truncated at 0.5/99.5% quantiles.

Figure 2: Boxplot showing the benefit of robustness under  $t(3)$ ,  $t(2.1)$ , and  $Pareto(2.5)$  tails

can deal with asymmetric distributions.

Catoni and MoR shows very similar results but, visually, MoR seems to slightly improve on Catoni's tails, but this conclusion is only drawn from a visual inspection of the graph. Even if Both estimator yielded the same simulation results, the parsimony of the MoR estimator may make it more appealing if the data is known to be heavy tailed.

### 7.3.3 $\delta$ -dependence

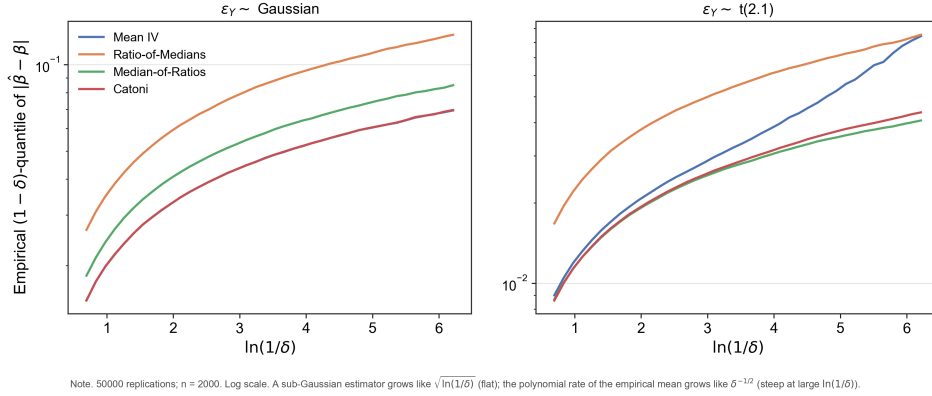


Figure 3: Graph showing the empirical  $(1 - \delta)$ -quantile of  $|\hat{\beta} - \beta|$  against  $\ln(1/\delta)$ . Catoni and Mean IV have a near identical graph in the Normal regime

Figure 3 shows that, for the Normal panel, All four curves are parallel and gently concave, meaning that every estimator is effectively sub-Gaussian, with Mean IV and Catonia sitting the lowest at every  $\delta$ . In the  $t(2,1)$  regime we can see that the Mean IV estimator becomes convex at around  $\ln(1/\delta) = 4$  which, as stated by Remark 3.4, is unavoidable. The more confidence is demanded, the more MoM wins when compared against Mean IV. The slight benefit that MoR showed against Catoni in Section 7.3.2 is also visible here, with MoR having a slightly smaller quantile than Catoni as  $\delta$  increases.

### 7.3.4 Instrument Strength Conditions

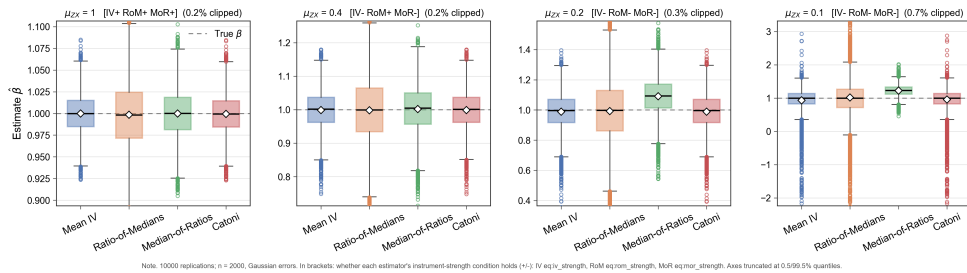


Figure 4: Boxplot showing the effect of weaker instrument strength conditions with a sweeping  $\mu_{ZX}$

Figure 4 shows that, at  $\mu_{ZX} = 0.4$ , where MoM conditions fail but IV holds, every estimator is still centered, but Mean IV and Catoni are the tightest. As the instrument strength condition weakens, IV and Catoni stay mean centered, but produce terrible estimates, with a standard

deviation of 2.5 and minimum and maximum values in the hundreds. MoR, though, remains stable, albeit highly biased, with a mean estimate of 1.23. On RMSE, MoR is the best fit here, with a value of 0.28, against 2.5 of Mean IV and 2.8 of Catoni. The RoM estimator performs worse on all cases. Under weak instruments, IV, RoM, and Catoni fail by variance explosion; and MoR fails by bias.

### 7.3.5 Empirical Size

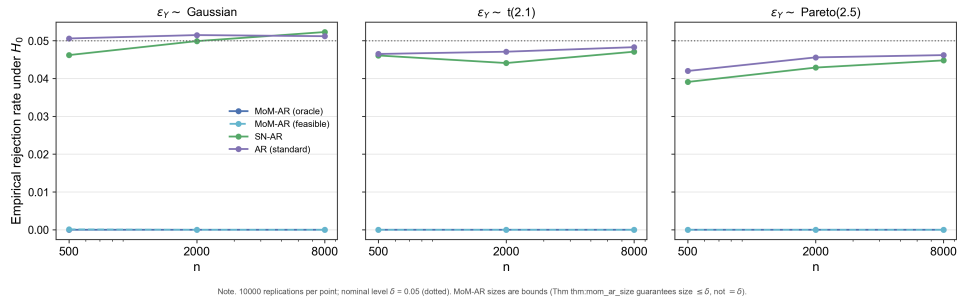


Figure 5: Plot of the size of all tests

Figure 5 shows that, at the true  $\beta$ , both the oracle and feasible MoM-AR test reject in at most 0.01% of 10,000 replications in every design, against the nominal 5%. The finite sample size bound of 6.1 holds and is very conservative; the oracle threshold  $\tau_n$  is 7.9 times the actual standard deviation of the statistic under  $H_0$  in the Gaussian design at  $n = 2000$ , and 16.7 times under  $t(2.1)$ ; it is built from Chebyshev and Hoeffding distribution agnostic worst case scenarios, so it is fixed by  $\sigma_{Z_\varepsilon}$  alone and cannot adapt to the tails. The feasible threshold, rebuilt from a robust scale estimate, instead stays between 7.0 and 7.3 times that standard deviation in every design and at every sample size.

Comparing the AR tests against each other, both are close enough to the nominal 5% size value, though as the distributions get heavier tails ( $t(2.1)$  and  $Pareto(2.5)$ ), they become slightly more conservative. It is worth noting that the self-normalised AR test is consistently slightly more conservative than the standard AR test.

### 7.3.6 Power of the Four Tests

Figure 6 shows that the MoM-AR acceptance regions are less powerful than the AR based tests, with the feasible version being slightly more powerful than the oracle version in the Normal setting. In the heavy tail setting, the feasible MoM-AR test is much more powerful. This is due to the MAD-based scale estimate setting  $\hat{\sigma}_{Z_\varepsilon}^{MAD}$  below the true  $\sigma_{Z_\varepsilon}$ . This is not a result that is expected and is a lucky cancellation due to the DGP, not a guarantee. The standard AR test is more powerful than the self-normalised AR test in the Normal setting, though this benefit decreases as tails become larger, as can be seen in the  $t(2.1)$  case, where the two are nearly indistinguishable. If the finite-sample guarantee is needed, one has to pay in power, though, as can be seen by comparing the self-normalised to the standard AR, the cost is not that high.

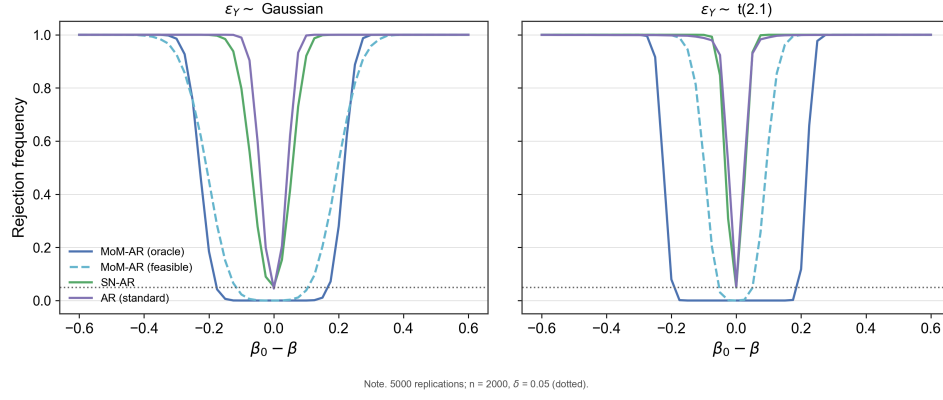


Figure 6: Power curves of all tests

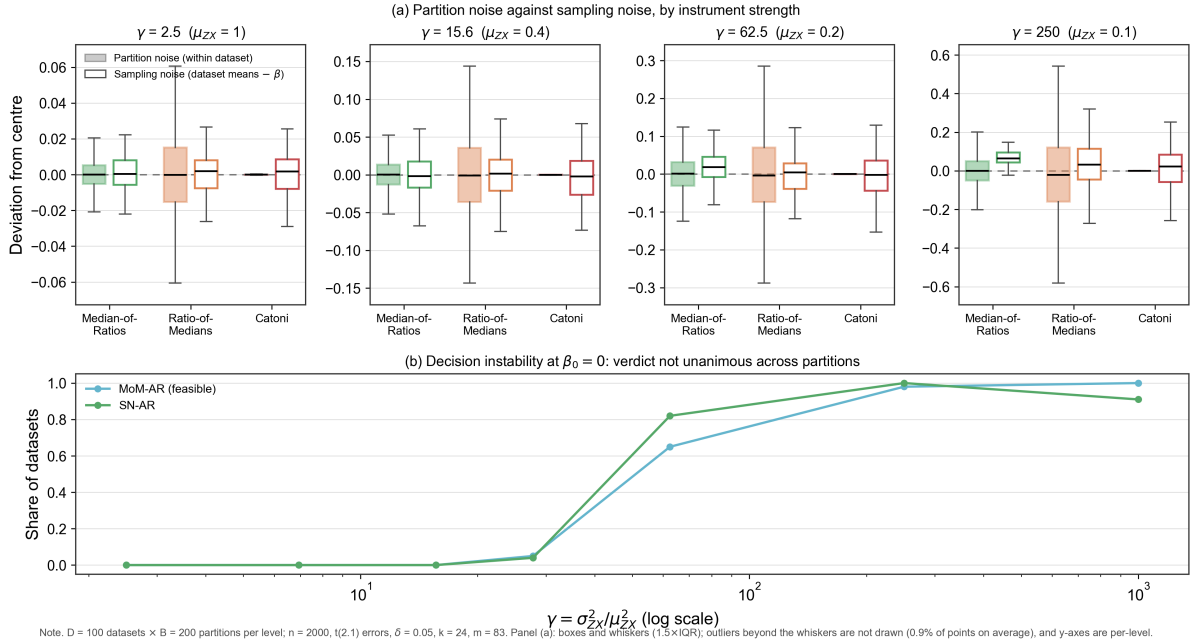


Figure 7: Partition noise against sampling noise, and the share of datasets whose test verdict at  $\beta_0 = 0$  depends on the partition, against instrument strength  $\gamma = \sigma_{ZX}^2 / \mu_{ZX}^2$

### 7.3.7 Partition Randomness

Every estimator of Sections 4–5 splits the sample into blocks, and on real data that split has to be randomised, since the observations do not arrive in a random order. The estimate therefore depends on the partition draw as well as on the data. Figure 7 separates the two. For each instrument strength,  $D = 100$  datasets are drawn and each is partitioned  $B = 200$  times. Panel (a) plots the deviation of every estimate from its own dataset’s average across partitions (filled boxes, which isolate the partition), against the dataset averages centred at the true  $\beta$  (hollow boxes, which isolate the data).

The partition is not a negligible source of noise even where the theory is comfortable. At  $\gamma = 2.5$ , where all strength conditions hold, the partition accounts for 34% of the total variance of the Median-of-Ratios estimator and for 78% of that of Ratio-of-Medians; in interquartile terms the Ratio-of-Medians partition box is roughly twice as wide as its sampling box. Catoni’s estimator is effectively partition-free, as the blocking enters it only through the variance pre-estimate. As the instrument weakens both block estimators deteriorate, and the deterioration is mostly in the partition: the Median-of-Ratios share rises to 0.93 at  $\gamma = 1000$ , so that at that point the reported number is more a function of the seed than of the sample.

Panel (b) asks what this costs a conclusion rather than an estimate. It reports the share of datasets on which the verdict at  $\beta_0 = 0$  is *not* unanimous across the 200 partitions, that is, on which rejecting the null depends on the seed. Up to  $\gamma \approx 16$  the verdict is unanimous on every dataset, so the partition is inferentially irrelevant. Instability then rises steeply, to 65% of datasets for the feasible MoM-AR test and 82% for the self-normalised test at  $\gamma = 62.5$ , and to essentially every dataset by  $\gamma = 250$ . The two tests behave alike, so this is a property of the blocking rather than of the threshold.

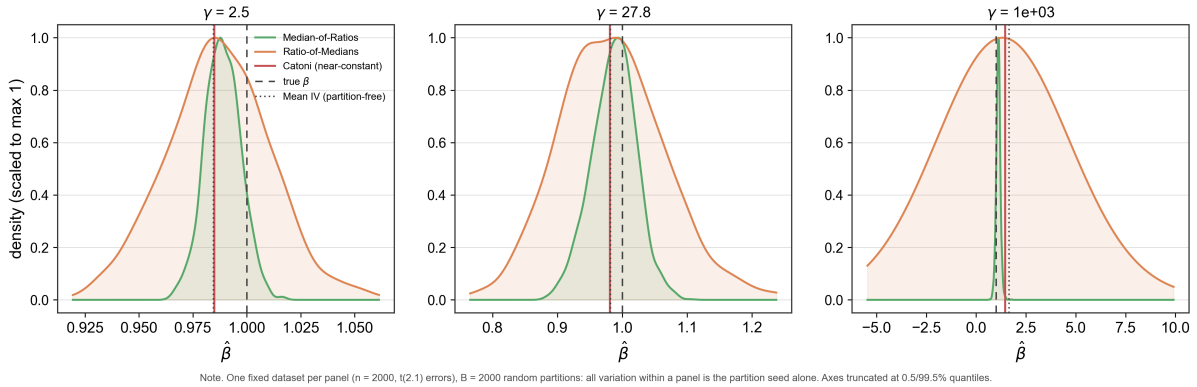


Figure 8: Estimator distributions over random partitions of a single fixed dataset, at three instrument strength levels

Figure 8 shows the same phenomenon from the position of a researcher holding one data set. Each panel fixes a single sample and redraws the partition 2,000 times, so all of the spread displayed is the seed alone; the partition-free Mean IV estimate is marked for reference. Under a strong instrument the two block estimators concentrate near it, with Ratio-of-Medians already visibly the wider of the two. At  $\gamma = 1000$  the contrast is stark: Ratio-of-Medians ranges over roughly  $[-5, 10]$  on a single data set, while Median-of-Ratios remains a narrow spike, displaced

from  $\beta$  but stable. This is the same division of failure modes seen in Section 7.3.4, and it appears here without any resampling of the data at all.

## 8 Case Studies

The simulations of Section 7.3 evaluated the estimators and tests on data generated to satisfy (A1)–(A3) by construction, with the instrument strength conditions holding or failing by design. This section applies the same procedures to two canonical instrumental variable studies, where neither the tail behaviour nor the instrument strength is under our control.

The two applications are chosen to sit on opposite sides of the conditions that drive the theory. Angrist and Krueger (1991) instrument years of schooling with quarter of birth; the textbook example of a weak instrument. There  $\mu_{ZX}$  is small relative to  $\sigma_{ZX}$  and every finite-sample strength condition of Sections 3–5 fails. Angrist and Evans (1998) instrument family size with the sex composition of the first two children. There instrument is stronger relative to the noise and, because both the regressor and the instrument are bounded indicators, all three strength conditions hold. Reading the two studies together therefore separates what the estimators do when the theory applies from what they do when it does not.

The purpose of this exercise is not to check whether the MoM-based estimators recover the published numbers, but rather to check what these methods buy, and cost, on real data.

expand

### 8.1 Empirical Strategy

#### 8.1.1 Mapping the applications into the scalar model

The theory developed concerns  $Y_i = \beta X_i + \varepsilon_i$  with a single instrument  $Z_i$  and  $\beta = \mu_{ZY}/\mu_{ZX}$ . Both applications are therefore taken in their Wald form, that is, the just-identified specification with a single binary instrument and no covariates. In each case the published paper reports this specification alongside its covariate-adjusted counterparts, which makes the replication target unambiguous: Panel B of Table III in Angrist and Krueger (1991) and columns (1)–(2) of Table 5 in Angrist and Evans (1998).

Both papers write their models with an intercept. Our methods have been developed without an intercept, so we demean the three variables at the full-sample level before applying any procedure,

$$Y_i \leftarrow Y_i - \bar{Y}, \quad X_i \leftarrow X_i - \bar{X}, \quad Z_i \leftarrow Z_i - \bar{Z}.$$

With a binary instrument the sample ratio  $\hat{\mu}_{ZY}/\hat{\mu}_{ZX}$  of the demeaned variables is then algebraically identical to the Wald estimator  $(\bar{Y}_1 - \bar{Y}_0)/(\bar{X}_1 - \bar{X}_0)$ , so the Mean IV estimator of Section 3 reproduces the published point estimate. Demeaning also keeps the moment  $W_i(\beta_0) = Z_i(Y_i - \beta_0 X_i)$  of Section 6.1 centered at zero under  $H_0$ , which would not hold if the intercept were left in.

The interpretation of the MoM-based estimators are the same as the Wald estimators; the estimands capture the local average treatment effect since they estimate the same ratio of population moments.

### 8.1.2 Blocking, and why the partition is randomised

Every procedure in Sections 4–6 partitions the sample into  $k$  blocks. In the simulations, the observations are exchangeable by construction, so any partition is as good as any other. The observations in Angrist and Krueger (1991) are not exchangeable; the data is sorted by year and quarter of birth. Thus, any blocks cut directly from the raw file would coincide with birth cohorts rather than with random subsamples, violating the i.i.i.d assumption. To solve this, we draw a uniformly random permutation of the rows and cut consecutive blocks from the permuted data, which assures our data is i.i.d.

This selection procedure presents a different issue. Conditional on the data, a different draw of the permutation gives a different estimate. Let the reported estimator be  $\hat{\beta}(\text{data}, \pi)$  for a random permutation  $\pi$ . The theorems developed bound the deviation for each fixed  $\pi$ , but the reported estimator

One solution could be to aggregate the results over multiple random permutations, but the probability statements developed in Sections 4–6 are taken over the sample. Therefore, we treat the permutation as an object of study in its own right, drawing  $B = 1,000$  independent partitions and reporting the resulting distribution of every partition-dependent quantity. Within a draw, all procedures share the same permutation, so differences across estimators are not confounded with differences in blocking.

The reported estimator then becomes  $\hat{\beta}(\text{data}, \pi)$  for a random permutation  $\pi$ . The theorems developed bound the deviation for a fixed  $\pi$

This is the methodological contribution of the section and deserves more than a paragraph. Expand with: (i) a formal statement that the reported estimator is  $\hat{\beta}(\text{data}, \pi)$  for a random permutation  $\pi$ , and that the theorems bound the deviation for *each fixed*  $\pi$ , so they transfer to the randomised estimator by averaging, but the width of the partition distribution is not controlled by them; (ii) the practical remedy, namely aggregating over partitions (median or mean of  $\hat{\beta}(\pi)$  over many draws), and whether the concentration guarantee survives aggregation; (iii) a note that the same issue affects any MoM estimator applied to ordered data, so it is not specific to the IV setting.

Decide and state a convention for the thesis: are the headline numbers one fixed seed, or aggregated over partitions? Currently the replication tables report a single documented seed and the sensitivity tables report the distribution. If aggregation is adopted, the tables must be regenerated.

### 8.1.3 Reported quantities

For each application we report, at  $\delta = 0.05$  throughout: the replication of the published table; the four point estimators (Mean IV, Ratio-of-Medians, Median-of-Ratios, and the Catoni competitor of Section 7.2); the three feasible confidence sets (MoM-AR with the robust scale estimate, the self-normalised AR test, and the standard AR test as a baseline); and a plug-in check of the finite-sample strength conditions. The oracle MoM-AR test of Theorem 6.1 is not feasible on real data, since  $\sigma_{Z_\varepsilon}$  is unknown, so only its feasible counterpart appears here.

Explain the plug-in strength check and its circularity: the conditions involve  $\mu_{ZX}$  and  $\sigma_{ZX}^2$ , which are replaced by sample moments, so the diagnostic is itself an estimate. Under a weak instrument  $\hat{\mu}_{ZX}$  is exactly the noisy quantity in question. Argue why the verdict is nonetheless informative (the failure margins are orders of magnitude, not borderline).

## 8.2 The Causal Effect of Compulsory School Attendance on Earnings

### 8.2.1 Data and replication

Angrist and Krueger (1991) exploit the interaction of compulsory schooling laws with school-entry rules: children born earlier in the calendar year reach the legal dropout age having completed less schooling, so quarter of birth shifts educational attainment without, the authors argue, affecting earnings through any other channel. We replicate Panel B of their Table III, the 1980 Census sample of men born 1930–1939, with  $Y$  the log weekly wage,  $X$  years of education and  $Z$  an indicator for birth in the first quarter.

Table 1 reproduces the published group means, differences and Wald estimate. The posted extract contains 329,509 observations against the 327,509 reported in the paper, as it is the sample used for their Table V; every reported quantity nonetheless matches to the precision printed in the original, with a Wald return to education of 0.1020 against a published 0.1020 and an OLS return of 0.0709 against 0.0709.

Comment on the economics: the Wald estimate exceeds OLS, which Angrist and Krueger (1991) read as evidence against simple ability bias. This is the substantive claim the confidence sets below are asked to adjudicate, so it should be stated before the results.

Document the two-thousand-observation discrepancy properly, and say whether it matters: it does not shift any replicated number at printed precision, but a reader will ask.

Table 1: Replication of Angrist and Krueger (1991), Table III, Panel B (1980 Census, men born 1930–1939). The posted extract has  $n = 329,509$  observations against the published 327,509; standard errors in parentheses.

|                           | Replicated |           | Published |           |
|---------------------------|------------|-----------|-----------|-----------|
|                           | Estimate   | (s.e.)    | Estimate  | (s.e.)    |
| ln(wkly. wage), born Q1   | 5.8916     |           | 5.8916    |           |
| ln(wkly. wage), born Q2–4 | 5.9027     |           | 5.9027    |           |
| difference                | -0.01110   | (0.00274) | -0.01110  | (0.00274) |
| Education, born Q1        | 12.6881    |           | 12.6881   |           |
| Education, born Q2–4      | 12.7969    |           | 12.7969   |           |
| difference                | -0.1088    | (0.0133)  | -0.1088   | (0.0132)  |
| Wald return to education  | 0.1020     | (0.0240)  | 0.1020    | (0.0239)  |
| OLS return to education   | 0.0709     | (0.00034) | 0.0709    | (0.0003)  |

### 8.2.2 Instrument strength

The plug-in moments of the demeaned data are  $\hat{\mu}_{ZX} = -0.0203$  and  $\hat{\sigma}_{ZX}^2 = 2.031$ , so  $\hat{\sigma}_{ZX}^2/\hat{\mu}_{ZX}^2 \approx 4,900$ . Every finite-sample condition of the point estimators fails, and by wide margins: the

Table 2: Robust IV estimates and 95% confidence sets for the return to education, AK91 extract ( $\delta = 0.05$ ; demeaned  $Y$ ,  $X$ ,  $Z$ ;  $k = 24$ ,  $m = 13,729$ ). The MoM-AR test uses the feasible robust scale estimate  $\hat{\sigma}_{Z\varepsilon} = 0.3429$ ; the oracle version is infeasible on real data.

| Estimator                                 | $\hat{\beta}$                                                   |
|-------------------------------------------|-----------------------------------------------------------------|
| Wald / Mean IV                            | 0.1020 (0.0240)                                                 |
| Ratio-of-Medians ( $k = 30$ )             | 0.1023                                                          |
| Median-of-Ratios ( $k = 24$ )             | 0.0584                                                          |
| Catoni ratio ( $\delta/2$ per coordinate) | 0.1018                                                          |
| Test                                      | 95% confidence set                                              |
| MoM-AR (feasible)                         | $[-0.1923, 0.3839]$                                             |
| SN-AR                                     | $[-0.0097, 0.1744] \cup [0.1761, 0.1935] \cup [0.2221, 0.2325]$ |
| AR (standard)                             | $[0.0554, 0.1525]$                                              |

standard IV condition (7) requires  $n \geq 789,667$  against the 329,509 available; the Median-of-Ratios condition (17) requires a block size  $m \geq 157,933$  against  $m = 13,729$ ; and the Ratio-of-Medians condition (14) requires  $m > 19,742$  against  $m = 10,983$ . The single-interval condition (31) fails by two orders of magnitude.

This is the empirical counterpart of experiment E3 (Section 7.3.4): the regime in which the guarantees are void and the estimators are used outside their warranted range. It is also the honest description of quarter of birth as an instrument.

Connect explicitly to the weak-instrument literature: the first-stage  $F$  statistic here, the Dufour (1997) argument for test inversion, and the observation that this is precisely the setting in which Wald-type confidence intervals are invalid, which motivates reporting the AR sets rather than  $\hat{\beta} \pm 1.96 \hat{\text{se}}$ . Compute and report the first-stage  $F$  for comparability with that literature.

### 8.2.3 Robust estimates and confidence sets

Table 2 reports the four estimators and the three confidence sets for one documented partition.

Write the reading of the table. Points that must appear: (i) Mean IV, Catoni and Ratio-of-Medians agree closely while Median-of-Ratios sits well below them at this seed, and Section 8.2.4 shows this is partition noise rather than a property of the estimator; (ii) the feasible MoM-AR set is far wider than the standard AR set and fails to exclude zero, consistent with the conservativeness of  $\tau_n$  documented in Section 7.3.5, where the threshold was roughly five times the actual standard deviation of the statistic; (iii) the self-normalised set is competitive with the standard AR set, and at this seed is a union of three intervals, an instance of the geometry of Corollary 6.6 that the simulations never triggered; (iv) none of the tests rejects the OLS value, so the headline finding of the paper is not overturned, but neither is it confirmed with the finite-sample machinery.

### 8.2.4 Partition sensitivity

Table 3 and Figure 9 summarise  $B = 1,000$  independent partitions of the same data.

The dispersion induced by the partition is of the same order as the sampling uncertainty of the Wald estimator: the Median-of-Ratios estimate has a standard deviation of 0.019 across partitions and ranges over  $[0.028, 0.150]$ , against a Wald standard error of 0.024. Catoni’s estimator is effectively partition-free, as it depends on the blocking only through its variance pre-estimate. The block means  $\hat{\mu}_{ZX}^{(j)}$  share a sign in only 30.4% of partitions, so the sufficient condition of Proposition 6.7 usually fails, and the self-normalised confidence set is a union of several intervals in 14.6% of draws. The feasible MoM-AR set excludes zero in 1.8% of partitions against 91.6% for the self-normalised set and 100% for the partition-free standard AR test.

Interpret rather than merely report. The key claim to defend: in a regime where the strength conditions fail, the choice of partition can change a qualitative inferential conclusion, and this is a cost of MoM-type procedures that the concentration theorems do not price. Contrast with Section 8.3.4, where the same diagnostics are benign.

The 10.1% rejection rate of the OLS value by the self-normalised test deserves comment, since the ability-bias conclusion of Angrist and Krueger (1991) is exactly the hypothesis at stake.

Table 3: Partition sensitivity on the AK91 extract:  $B = 1,000$  independent random partitions, each shared by all MoM-based procedures ( $\delta = 0.05$ ). The partition-free Wald estimate is 0.1020. Top panel: dispersion of the point estimates across partitions. Bottom panel: rejection and shape frequencies of the 95% confidence sets; the standard AR test does not depend on the partition and is shown for reference.

| Estimator                     | Mean               | SD                           | Q <sub>0.05</sub> | Q <sub>0.95</sub> | Range              |
|-------------------------------|--------------------|------------------------------|-------------------|-------------------|--------------------|
| Median-of-Ratios              | 0.0982             | 0.0189                       | 0.0666            | 0.1284            | $[0.0280, 0.1496]$ |
| Ratio-of-Medians              | 0.1016             | 0.0211                       | 0.0686            | 0.1373            | $[0.0409, 0.2036]$ |
| Catoni ratio                  | 0.1018             | 0.0000                       | 0.1018            | 0.1019            | $[0.1018, 0.1019]$ |
| Test                          | % rej. $\beta_0=0$ | % rej. $\beta_0=\beta_{OLS}$ | % split           | % unbounded       | Med. length        |
| MoM-AR (feasible)             | 1.8                | 0.0                          | 0.1               | 0.0               | 0.431              |
| SN-AR                         | 91.6               | 10.1                         | 14.6              | 0.0               | 0.145              |
| AR (standard, partition-free) | 100.0              | 0.0                          | 0.0               | 0.0               | 0.097              |

## 8.3 The Causal Effect of Family Size on Parental Labour Supply

### 8.3.1 Data and replication

Angrist and Evans (1998) estimate the effect of childbearing on parents’ labour supply, instrumenting an indicator for having more than two children with an indicator for the first two children having the same sex. Parents with two same-sex children are more likely to have a third, while the sex mix itself is as good as randomly assigned, which is the identifying claim. We replicate columns (1)–(2) of their Table 5: the 1980 Census sample of all women aged 21–35 with two or more children, for the five labour-supply outcomes.

Table 4 reports the replication. The reconstructed sample contains 394,840 observations against the 394,835 of the published table, a five-observation discrepancy of the posted archive also reported by Farbmacher, Guber and Vikström (2018), whose selection rules we follow. The

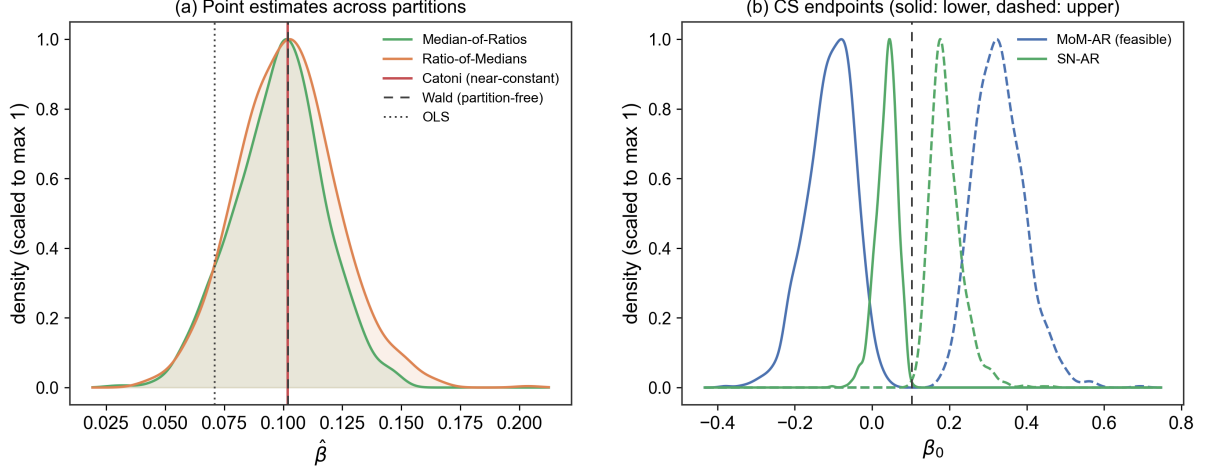


Figure 9: Partition sensitivity on the Angrist and Krueger (1991) extract over  $B = 1,000$  random partitions. Panel (a): kernel density estimates of the point estimates, each scaled to unit maximum, with the partition-free Wald and OLS values marked. Panel (b): densities of the outer endpoints of the 95% confidence sets (solid: lower, dashed: upper).

first stage is 0.0595 against a published 0.0600, and all five Wald estimates match at printed precision.

Give the sample construction in the text or an appendix: age 21–35, two or more children, second child older than one year, mother at least 15 at first birth, and no Census-allocated values on the variables used. State that the published data appendix is in the working-paper version and that the rules are taken from Farbmacher et al. (2018), who work from the same file.

Explain why the 1980 rather than the 1990 sample is used: the 1990 results require PUMS person weights, which take the analysis outside the i.i.d. framework of (A1)–(A3), and twins are mismeasured in 1990 without quarter of birth.

### 8.3.2 Instrument strength

Here  $\hat{\mu}_{ZX} = 0.0149$  and  $\hat{\sigma}_{ZX}^2 = 0.0598$ , so  $\hat{\sigma}_{ZX}^2/\hat{\mu}_{ZX}^2 \approx 270$ , roughly eighteen times smaller than in Section 8.2.2. The standard IV condition (7) requires  $n \geq 43,219$  and is met with 394,840 observations; the Median-of-Ratios condition (17) requires  $m \geq 8,644$  and is met with  $m = 16,451$ ; the Ratio-of-Medians condition (14) requires  $m > 1,080$  and is met with  $m = 13,161$ . Only the single-interval condition (31), the most demanding of the four, fails.

Explain *why* the conditions hold here and not in Section 8.2. The mechanism is that both  $X$  and  $Z$  are bounded indicators, so  $Z_i X_i \in \{0, 1\}$  up to centring and  $\sigma_{ZX}^2$  is mechanically small, whereas years of schooling has a large variance. This is a substantive point about when the conditions are attainable in practice and should not be left implicit.

Table 4: Replication of Angrist and Evans (1998), Table 5, columns (1)–(2): 1980 PUMS, all women aged 21–35 with two or more children, *Same sex* instrument, *More than 2 children* as endogenous regressor. Reconstructed sample  $n = 394,840$  against the published 394,835 (five-observation archive discrepancy documented by Farbmacher et al. (2018)). Standard errors in parentheses; Wald SEs by the delta method.

|                               | Replicated |          | Published |          |
|-------------------------------|------------|----------|-----------|----------|
|                               | Estimate   | (s.e.)   | Estimate  | (s.e.)   |
| <i>More than 2 children</i>   | 0.0595     | (0.0016) | 0.0600    | (0.0016) |
| <i>Number of children</i>     | 0.0754     | (0.0026) | 0.0765    | (0.0026) |
| Worked for pay, difference    | -0.0079    | (0.0016) | -0.008    | (0.0016) |
| Wald                          | -0.132     | (0.026)  | -0.133    | (0.026)  |
| Weeks worked, difference      | -0.3787    | (0.0709) | -0.3826   | (0.0709) |
| Wald                          | -6.360     | (1.181)  | -6.38     | (1.17)   |
| Hours/week, difference        | -0.3089    | (0.0602) | -0.311    | (0.0602) |
| Wald                          | -5.187     | (1.006)  | -5.18     | (1.0)    |
| Labor income, difference      | -130.3     | (34.4)   | -132.5    | (34.4)   |
| Wald                          | -2188.3    | (573.6)  | -2208.8   | (569.2)  |
| ln(Family income), difference | -0.0016    | (0.0041) | -0.0018   | (0.0041) |
| Wald                          | -0.026     | (0.069)  | -0.029    | (0.068)  |

Table 5: Robust IV estimates and 95% confidence sets, AE98 1980 all-women sample ( $\delta = 0.05$ ; demeaned  $Y$ ,  $X$ ,  $Z$ ; one shared random partition,  $k = 24$ ,  $m = 16,451$  for MoR and the tests,  $k = 30$  for RoM). The MoM-AR test uses the feasible robust scale estimate; the oracle version is infeasible on real data.

| Outcome           | Wald/Mean IV            | MoR                     | RoM                      | Catoni  |
|-------------------|-------------------------|-------------------------|--------------------------|---------|
| Worked for pay    | -0.1318                 | -0.1118                 | -0.1398                  | -0.1318 |
| Weeks worked      | -6.3596                 | -6.2532                 | -7.7001                  | -6.3595 |
| Hours/week        | -5.1872                 | -4.8504                 | -6.6122                  | -5.1871 |
| Labor income      | -2188.3                 | -1734.1                 | -2860.9                  | -2188.4 |
| ln(Family income) | -0.0262                 | -0.0436                 | 0.0481                   | -0.0262 |
| Outcome           | MoM-AR (feasible)       | SN-AR                   | AR (standard)            |         |
| Worked for pay    | [-0.3074, 0.0725]       | [-0.1703, -0.0513]      | [-0.1835, -0.0802]       |         |
| Weeks worked      | [-18.4603, 5.4799]      | [-10.3211, -2.4699]     | [-8.6785, -4.0427]       |         |
| Hours/week        | [-14.8783, 3.7690]      | [-8.7151, -2.4946]      | [-7.1639, -3.2160]       |         |
| Labor income      | [-5012.5018, 1281.2594] | [-2763.7360, -683.3022] | [-3312.9510, -1061.4915] |         |
| ln(Family income) | [-1.1038, 1.1935]       | [-0.3623, 0.2625]       | [-0.1609, 0.1094]        |         |

### 8.3.3 Robust estimates and confidence sets

Write the reading of Table 5. Points to cover: (i) all four estimators agree closely for every outcome, as the theory predicts when the strength conditions hold and the tails are moderate; (ii) the feasible MoM-AR sets are again roughly four times longer than the standard AR sets and never exclude zero, so the conservativeness of  $\tau_n$ , not instrument strength, is the binding constraint; (iii) the self-normalised sets exclude zero for the four labour-supply outcomes and are close to the standard AR sets; (iv)  $\ln(\text{Family income})$  is the interesting case: no test excludes zero, matching the paper's own null result, while the standard AR test excludes the OLS value and the self-normalised test does not.

### 8.3.4 Partition sensitivity

Write this subsection around Table 6 and Figures 10 and 11. The findings to report: across all five outcomes and  $B = 1,000$  partitions, the estimator distributions are centred on the partition-free Wald value; the partition standard deviation is roughly 70% of the sampling standard error (for *Worked for pay*, 0.018 against 0.026); and no confidence set, for any outcome or any partition, is split or unbounded, against 14.6% split sets in Section 8.2.4. Rejection frequencies are stable rather than knife-edge: the self-normalised test excludes zero in 99% or more of partitions for the three labour-supply outcomes and 90.4% for labour income.

Draw the comparative conclusion, which is the payoff of running two case studies: partition randomness is a second-order nuisance when the strength conditions hold and a first-order threat when they do not. Note that the diagnostic is available to a practitioner without knowing the truth, since both the strength check and the partition study are computable from the data.

Decide how many of the five sensitivity figures to include. Suggestion: keep *Worked for pay* and *Labor income* in the body, the first as the headline binary outcome and the second as the heavy-tailed one, and move the remaining three to the appendix.

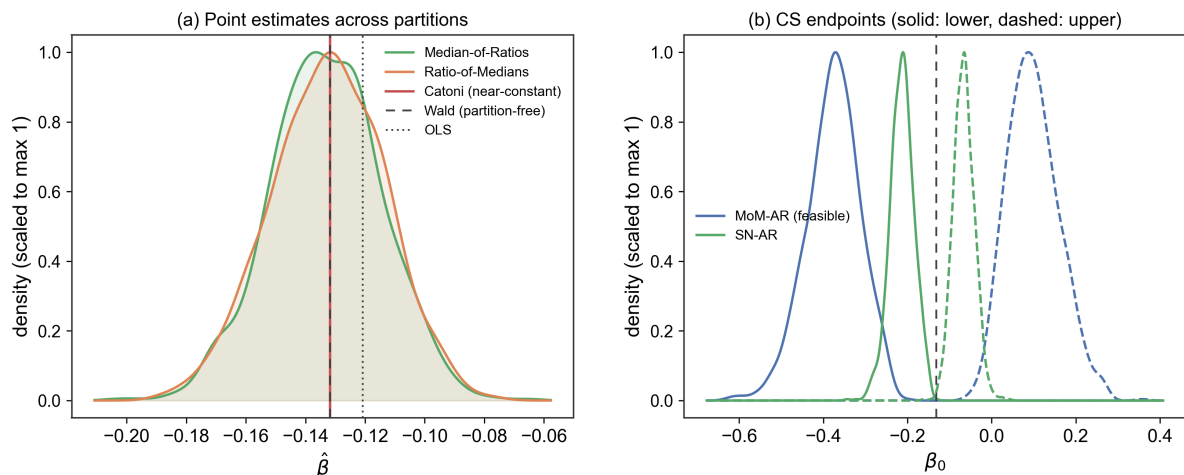


Figure 10: Partition sensitivity for *Worked for pay*, Angrist and Evans (1998) 1980 all-women sample,  $B = 1,000$  random partitions. Panels as in Figure 9.

Table 6: Partition sensitivity on the AE98 1980 all-women sample:  $B = 1,000$  random partitions shared by all MoM-based procedures and reused across outcomes ( $\delta = 0.05$ ). For each outcome, the top rows give the dispersion of the partition-dependent point estimates (the partition-free Wald estimate in the label); the bottom rows give rejection and shape frequencies of the 95% confidence sets. The standard AR test is partition-free and shown for reference.

| Outcome                          | Estimator                     | Mean               | SD                           | Q <sub>0.05</sub> | Q <sub>0.95</sub> |
|----------------------------------|-------------------------------|--------------------|------------------------------|-------------------|-------------------|
| Worked for pay (Wald -0.1318)    | Median-of-Ratios              | -0.1331            | 0.0184                       | -0.1641           | -0.1028           |
|                                  | Ratio-of-Medians              | -0.1325            | 0.0189                       | -0.1638           | -0.1016           |
|                                  | Catoni ratio                  | -0.1318            | 0.0000                       | -0.1318           | -0.1318           |
| Weeks worked (Wald -6.3596)      | Median-of-Ratios              | -6.4039            | 0.8206                       | -7.7276           | -5.0832           |
|                                  | Ratio-of-Medians              | -6.3876            | 0.8753                       | -7.8407           | -4.9402           |
|                                  | Catoni ratio                  | -6.3595            | 0.0000                       | -6.3595           | -6.3595           |
| Hours/week (Wald -5.1872)        | Median-of-Ratios              | -5.1964            | 0.6893                       | -6.3345           | -4.0640           |
|                                  | Ratio-of-Medians              | -5.2486            | 0.7021                       | -6.4069           | -4.0759           |
|                                  | Catoni ratio                  | -5.1871            | 0.0000                       | -5.1871           | -5.1871           |
| Labor income (Wald -2188.3)      | Median-of-Ratios              | -2195.0            | 395.1                        | -2833.3           | -1546.7           |
|                                  | Ratio-of-Medians              | -2170.3            | 398.9                        | -2823.9           | -1526.5           |
|                                  | Catoni ratio                  | -2188.4            | 0.0                          | -2188.4           | -2188.4           |
| ln(Family income) (Wald -0.0262) | Median-of-Ratios              | -0.0257            | 0.0469                       | -0.1032           | 0.0497            |
|                                  | Ratio-of-Medians              | -0.0249            | 0.0487                       | -0.1032           | 0.0559            |
|                                  | Catoni ratio                  | -0.0262            | 0.0000                       | -0.0262           | -0.0262           |
| Outcome                          | Test                          | % rej. $\beta_0=0$ | % rej. $\beta_0=\beta_{OLS}$ | % split           | % unb.            |
| Worked for pay                   | MoM-AR (feasible)             | 3.5                | 0.0                          | 0.0               | 0.0               |
|                                  | SN-AR                         | 99.3               | 1.2                          | 0.0               | 0.0               |
|                                  | AR (standard, partition-free) | 100.0              | 0.0                          | 0.0               | 0.0               |
| Weeks worked                     | MoM-AR (feasible)             | 7.5                | 0.0                          | 0.0               | 0.0               |
|                                  | SN-AR                         | 99.6               | 1.4                          | 0.0               | 0.0               |
|                                  | AR (standard, partition-free) | 100.0              | 0.0                          | 0.0               | 0.0               |
| Hours/week                       | MoM-AR (feasible)             | 6.0                | 0.0                          | 0.0               | 0.0               |
|                                  | SN-AR                         | 99.0               | 5.9                          | 0.0               | 0.0               |
|                                  | AR (standard, partition-free) | 100.0              | 0.0                          | 0.0               | 0.0               |
| Labor income                     | MoM-AR (feasible)             | 1.1                | 0.0                          | 0.0               | 0.0               |
|                                  | SN-AR                         | 90.4               | 1.2                          | 0.0               | 0.0               |
|                                  | AR (standard, partition-free) | 100.0              | 0.0                          | 0.0               | 0.0               |
| ln(Family income)                | MoM-AR (feasible)             | 0.0                | 0.0                          | 0.0               | 0.0               |
|                                  | SN-AR                         | 1.2                | 18.2                         | 0.0               | 0.0               |
|                                  | AR (standard, partition-free) | 0.0                | 100.0                        | 0.0               | 0.0               |

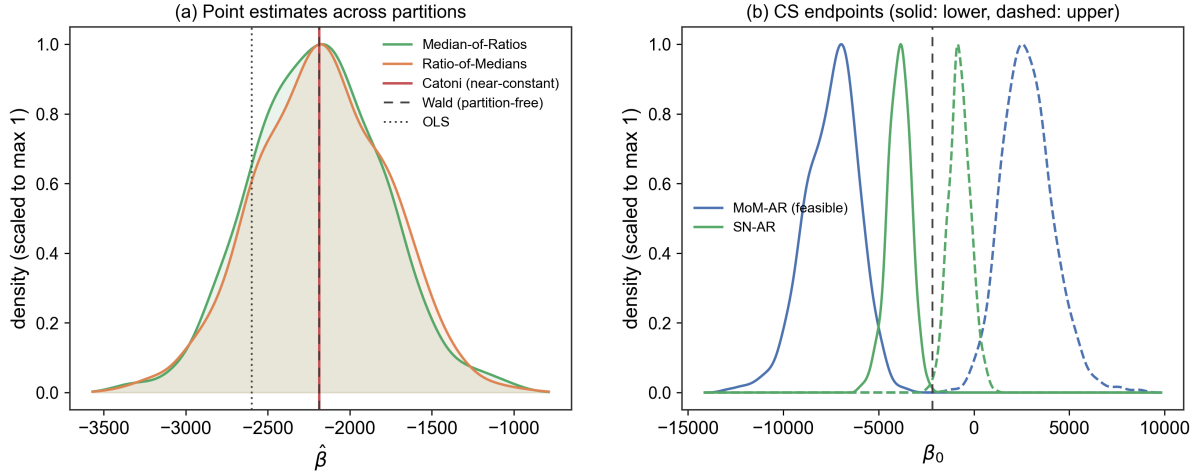


Figure 11: Partition sensitivity for *Labor income*, Angrist and Evans (1998) 1980 all-women sample,  $B = 1,000$  random partitions. Panels as in Figure 9.

Add, or move to the appendix, the remaining three figures:  
`ae98_seed_sensitivity_weeksm.png`, `_hoursm.png` and `_lfaminc.png`, following the label  
convention `fig:ae98_sensitivity_<outcome>`.

## 8.4 Discussion

### 8.4.1 Do these data have the tails the theory is built for?

The estimators of this thesis are designed for heavy-tailed  $Z\varepsilon$ , and Section 7.3.2 shows their advantage materialising only once the tails are heavy. Whether the advantage is available in an application is therefore an empirical question about the data, and it should be asked before the results are interpreted.

The sample kurtosis of  $Z_i\hat{\varepsilon}_i$ , evaluated at the Wald estimate, is 44 for the Angrist and Krueger (1991) log wage equation, against 3 for a Gaussian. In Angrist and Evans (1998) it ranges from 1.1 for *Worked for pay* and 1.4 for *Weeks worked*, both lighter-tailed than a Gaussian because the outcome is bounded, to 26 for *Labor income* and 35 for  $\ln(\text{Family income})$ . The applications therefore span both regimes, and within the second application the tail behaviour varies by outcome while the instrument and regressor are held fixed.

These diagnostics are currently computed ad hoc and must be folded into the replication scripts so that the numbers regenerate with the tables. A tail-index estimate would be more informative than kurtosis, but the naive Hill estimator is degenerate for the bounded outcomes, where the top order statistics tie; either restrict it to the continuous outcomes or use an estimator that tolerates ties.

Then make the argument this section exists to make: the case studies do *not* show the MoM estimators beating the sample mean, and should not be presented as if they did. *Worked for pay* is bounded, so the sample mean is already sub-Gaussian and blocking can only discard information; the two heavy-tailed outcomes are the ones where a difference could appear. Investigate whether the partition-averaged MoR estimate has a visibly tighter distribution for *Labor income* than for *Worked for pay*, which is the only place in these data where the theory predicts a gain.

#### 8.4.2 What the case studies establish

Summarise, honestly and without overclaiming. A defensible list: (i) the estimators and tests are implementable at census scale and reproduce the published Wald estimates exactly, so nothing is broken; (ii) the finite-sample strength conditions are informative in practice, since they separate the two applications in the way the theory predicts and are checkable from the data; (iii) the finite-sample threshold  $\tau_n$  is too conservative to be useful for inference in either application, and the self-normalised test is the one that should be recommended; (iv) partition randomness is a real and previously unremarked cost of applying MoM procedures to ordered data.

#### 8.4.3 Limitations

Collect the limitations, several of which have been flagged above: no covariates, so the replications are the papers' simplest specifications and not their preferred ones; the scalar theory cannot accommodate the overidentified designs both papers ultimately use; full-sample demeaning induces  $O(1/n)$  cross-block dependence; the 1990 Angrist and Evans (1998) sample is excluded because it requires survey weights; top-coding of the income variables truncates exactly the tail the theory cares about, which may understate the case for robustness; and the plug-in strength diagnostics are estimates, not oracle statements.

Note the top-coding point carefully. Labor income is top-coded at \$75,000 in the 1980 Census, so the observed tail is censored and the measured kurtosis of 26 is a lower bound on the underlying tail heaviness. This cuts against the case studies as a test of the heavy-tail machinery and should be acknowledged rather than buried.

## 9 Appendix

### 9.1 Self-Normalised Anderson-Rubin Test table

Table 7: Simulated critical values  $c_{k,\delta}$ : the  $(1 - \delta)$  quantiles of  $R_k$  (Proposition 6.13).

| $k$ | $\delta = 0.1$ | $\delta = 0.05$ | $\delta = 0.025$ | $\delta = 0.01$ |
|-----|----------------|-----------------|------------------|-----------------|
| 5   | 2.773          | 4.236           | 6.275            | 10.300          |
| 10  | 1.569          | 2.085           | 2.672            | 3.614           |
| 15  | 0.938          | 1.181           | 1.438            | 1.798           |
| 19  | 0.800          | 0.996           | 1.190            | 1.459           |
| 20  | 0.846          | 1.052           | 1.259            | 1.540           |
| 24  | 0.743          | 0.916           | 1.086            | 1.311           |
| 25  | 0.673          | 0.828           | 0.980            | 1.177           |
| 30  | 0.640          | 0.784           | 0.920            | 1.100           |
| 37  | 0.535          | 0.651           | 0.761            | 0.902           |
| 40  | 0.535          | 0.649           | 0.758            | 0.895           |
| 50  | 0.469          | 0.567           | 0.659            | 0.772           |

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